

Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention

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Abstract

Multi-Agent Debate has emerged as a promising framework for improving the reasoning quality of large language models through iterative inter-agent communication. However, broadcasting all agent messages at every round introduces noise and redundancy that can degrade debate quality and waste computational resources. Current approaches rely on uncertainty estimation to filter low-confidence responses before broadcasting, but this approach is unreliable due to miscalibrated confidence scores and sensitivity to threshold selection. To address this, we propose **Diversity-Aware Retention (DAR)**, a lightweight debate framework that, at each debate round, selects the subset of agent responses that maximally disagree with each other and with the majority vote before broadcasting. Through an explicit index-based retention mechanism, DAR preserves the original messages without modification, ensuring that retained disagreements remain authentic. Experiments on diverse reasoning and question answering benchmarks demonstrate that our selective message propagation consistently improves debate performance, particularly as the number of agents scales, where noise accumulation is most severe. Our results highlight that what agents hear is as important as what agents say in multi-agent reasoning systems. Code is publicly available at <https://github.com/DA2I2-SLM/DAR>.

1 Introduction

Large-Language-Model-based Multi-Agent Systems (LLM-based MAS) have recently attracted increasing attention thanks to their ability to leverage the power of LLMs in modeling and coordinating multiple agents, enabling complex problem solving and distributed decision-making (Tran et al., 2025; Yan et al., 2025). Within this paradigm, Multi-Agent Debate (MAD) has emerged as a main direction for improving the performance of LLM-based MAS. By enabling multiple language model agents to iteratively exchange and critique each other’s responses, MAD has demonstrated consistent gains over single agents across a range of reasoning and question answering benchmarks (Chan et al., 2023; Du et al., 2024; Liang et al., 2024).

Despite its promise, MAD faces a fundamental scalability challenge: as the number of agents and debate rounds grows, each agent has to process an ever-expanding context consisting of all peer responses from the previous round. This broadcasting of all messages introduces two problems. First, redundant responses, where multiple agents converge on the same answer through similar reasoning paths, provide little additional signal while consuming significant token capacity. Second, noisy or low-quality generations can actively mislead other agents, propagating errors across rounds rather than correcting them. These effects reduce the practical benefits of scaling debate and raise questions about MAD (Choi et al., 2025), particularly regarding what information agents should be provided.

A simpler solution for this problem is to restrict inter-agent communication by using a fixed sparse topology (Li et al., 2024; Choi et al., 2025). However, this method fails to account for the varying relevance of messages, as a static topology cannot distinguish between critical and redundant information, potentially discarding useful context or propagating noise. Later work has sought to address message quality through uncertainty-based filtering, discarding responses whose confidence scores fall below a chosen threshold before broadcasting (Lin & Hooi, 2025; Tian et al., 2026). While intuitive, this approach has significant limitations in practice: confidence estimates from LLMs are often poorly calibrated (Kadavath et al., 2022;

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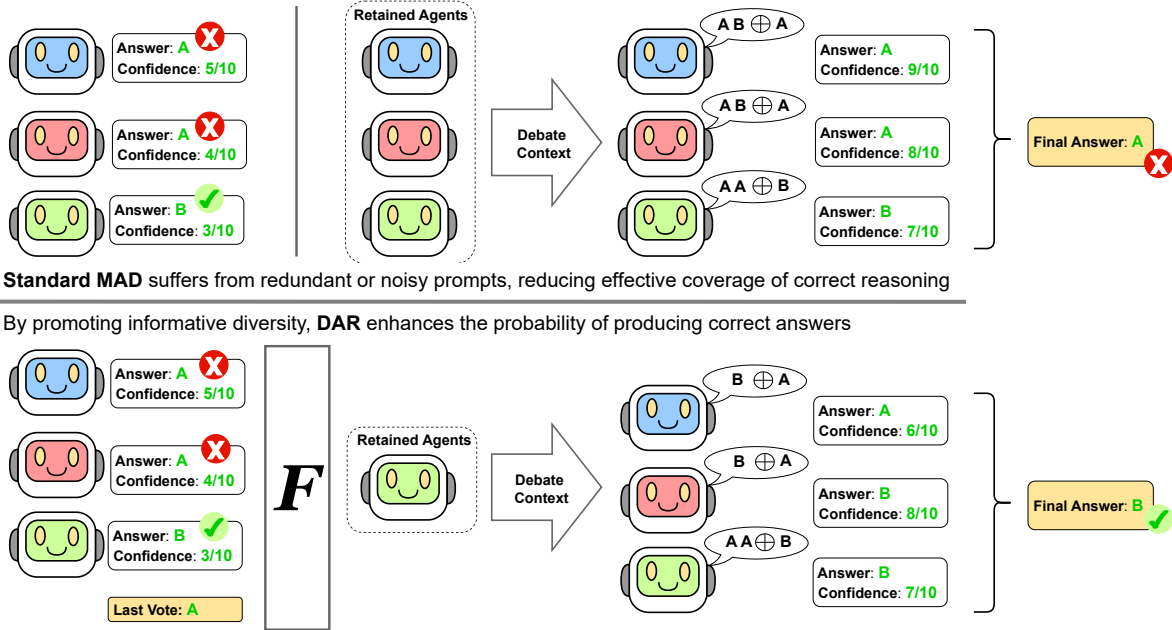


Figure 1: In standard MAD (top), each agent receives all peer responses (e.g., A, B) as context, which can be redundant and noisy. By the end of the debate round, the noisy message (A) outnumbers the correct one (B): 67% vs. 33%. In **DAR** (bottom), together with appending confidence and last vote result, we introduce the filter module F to remove redundancy and maintain disagreement across debate rounds, increasing the chance of generating correct answers even when they start as the minority (A vs. B before filtering: 67% vs. 33%; after filtering: 57% vs. 43%).

Xiong et al., 2024), and performance is highly sensitive to the choice of threshold, requiring costly tuning that may not generalize across models or tasks. Critically, all of these methods ignore a key property of productive debate: disagreement. A high-confidence response that merely echoes the majority view contributes far less to collective reasoning than a lower-confidence response that suggests an alternative perspective.

We argue that capturing disagreement among agent responses is central to effective debate, as diverse reasoning paths are more informative than redundant ones. To this end, we propose **Diversity-Aware Retention (DAR)**, a MAD framework that, at each debate round, selects the subset of agents' high-quality responses that maximally disagree with one another and with the majority vote (Figure 1). Concretely, DAR augments the standard debate protocol with three lightweight components. **First**, each agent's prompt is augmented with an uncertainty score for every peer response, computed as the average negative log-likelihood over the answer tokens, allowing agents to weight peer responses according to their reliability. By introducing the uncertainty score into the prompt as additional information for the agent's generation, we avoid the challenge of tuning the confidence threshold. **Second**, DAR prepends the majority vote from the previous round to each agent's prompt, providing an explicit consensus anchor across debating rounds, which will be utilized in the next step. **Third** and most importantly, we introduce a filter agent that receives all peer responses and the prepended majority vote, then outputs a set of agent indices identifying the most mutually disagreeing responses. The original messages at those indices are retrieved and broadcast unchanged. This index-based selection guarantees that the retained disagreements are original, enabling a controlled intervention in the debate process. Remarkably, our method is agnostic to the underlying communication topology, supporting both fully connected and sparse structures. We conduct extensive experiments on six diverse reasoning and question answering benchmarks across four LLM backbones ranging from 1.5B to 8B. Our results demonstrate that DAR consistently improves debate performance, particularly as the number of agents scales, while reducing the computational cost of message broadcasting.

2 Related Work

Multi-Agent Debate. MAD was first introduced by Du et al. (2024), improving response quality through simulated debates among multiple LLM agents. Extensions include assigning diverse roles (Liang et al., 2024; Chan et al., 2023), introducing mechanisms such as diversity pruning and misunderstanding refutation (Estornell & Liu, 2024), and designing more effective communication protocols (Li et al., 2024; Pham et al., 2024; Liu et al., 2023; 2024). Recent work has explored richer signals to improve debate dynamics, such as incorporating explicit confidence expression to guide interactions and mitigate premature convergence (Lin & Hooi, 2025). In parallel, ensemble-style multi-agent frameworks like *Tool-Use Mixture (TUMIX)* leverage diverse tool-use strategies with iterative refinement across agents, emphasizing the role of diversity and coordination in improving reasoning performance (Chen et al., 2026). Despite these advances, MAD is sensitive to hyperparameters and does not always outperform strong single-agent reasoning (Smit et al., 2024). Debates may converge prematurely, and simple majority voting can achieve similar results (Estornell & Liu, 2024; Choi et al., 2025), suggesting that improvements often come from candidate generation rather than debate itself. Recent work addresses this by filtering information between rounds, for example, *subjective masking* uses an LLM agent to self-evaluate each response, while *objective masking* discards low-confidence responses (Tian et al., 2026). Another approach increase candidate diversity through more aggressive sampling, albeit at higher computational cost (Zhu et al., 2026). In contrast, our method does not rely on *per-sample* judgments or expensive sampling. The proposed filtering module retains responses that disagree with the majority vote, while ensuring that the selected subset preserves diverse viewpoints. This enhances MAD performance by preserving informative diversity in a simple and computationally efficient manner.

Uncertainty and Diversity in Multi-Agent LLM Systems. Uncertainty estimation has been widely studied as a proxy for reliability in LLMs. Prior work measures uncertainty using token-level probabilities (Guerreiro et al., 2023; Manakul et al., 2023; Nguyen et al., 2026), verbalized confidence (Lin et al., 2022), or semantic entropy (Kuhn et al., 2023). More recent approaches extend this to multi-step settings, such as propagating uncertainty across reasoning steps to better capture cumulative errors in agent-based systems (Zhao et al., 2025). In multi-agent debate, incorporating confidence signals has been shown to improve aggregation and reduce premature convergence (Lin & Hooi, 2025). Orthogonally, diversity has been identified as a key factor in improving reasoning performance. Prior work shows that gains in multi-agent systems often arise from diverse candidate generation rather than interaction itself (Smit et al., 2024; Wu et al., 2025). Methods such as self-consistency, prompt diversification, and multi-prompt ensembles explicitly encourage multiple reasoning paths via parallel prompting, leading to more robust aggregation (Wang et al., 2022; Hu et al., 2025). These observations suggest a close connection between uncertainty and diversity, where uncertainty can help identify informative or complementary candidates. However, existing approaches typically rely on explicit uncertainty calibration or increased sampling cost, motivating lightweight alternatives that leverage implicit signals, such as disagreement, to preserve informative diversity without explicit uncertainty modeling.

3 Method

3.1 Preliminaries

Consider a debate with N ($N \geq 2$) agents $\{a_1, \dots, a_N\}$ over R ($R \geq 1$) rounds. Let $g_{r,i}$ denote the generation of agent a_i at round r . The debate state at each round consists of the previous-round responses $G_{r-1} = \{g_{r-1,1}, \dots, g_{r-1,N}\}$. This captures the standard multi-agent debate setting, where each agent conditions its response on the collective outputs of its peers via shared context aggregation, which may be implemented through concatenation or more structured strategies.

3.2 Incorporating Uncertainty in Multi-Agent Debate

Recent work has shown that confidence-aware debate improves multi-agent reasoning by enabling calibrated communication (Lin & Hooi, 2025). We compute uncertainty using the average negative log-likelihood (ANLL) over answer tokens (Huang et al., 2025):

$$\text{ANLL}(a) = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(a^t \mid a^{<t}, s), \quad (1)$$

where a^t is the t -th token in a (length T) and p_θ is the model distribution. These scores are appended to each agent’s prompt as additional cues, extending standard MAD with uncertainty signals without altering the debate protocol. The prompt template is provided in Appendix A.3.

3.3 Majority Vote as Additional Context

Large language models are known to benefit from richer in-context information, as demonstrated in few-shot and in-context learning settings (Brown et al., 2020; Min et al., 2022). Inspired by the observation of Choi et al. (2025) that majority voting alone can match the performance of full debate, we introduce a *soft* integration of the majority signal into the debate process. Formally, let $\text{ans}(g)$ denote the final answer extracted from a generation g . The majority-vote answer at round $r-1$, referred to as the *last vote*, is defined as:

$$v_{r-1} = \text{mode} \left(\{\text{ans}(g_{r-1,i})\}_{i=1}^N \right), \quad (2)$$

i.e., the most frequent answer among the agents’ responses in round $r-1$. At round r , we incorporate this signal by prepending the last vote v_{r-1} to the prompt of each agent, together with the responses from the previous round G_{r-1} (See Appendix A.3 for the prompt template).

Providing the last vote serves as an explicit consensus anchor, reducing the need to infer agreement patterns, which can be challenging for smaller models (Touvron et al., 2023; Riviere et al., 2024). It guides the next-round generation in two ways: (1) reinforcing consensus by encouraging verification against supporting evidence, and (2) highlighting disagreements to promote focused refinement. This improves context utilization while preserving the original debate dynamics¹.

3.4 Improving MAD by Promoting Diversity

While MAD benefits from aggregating multiple candidate responses, not all generations are equally useful. LLM outputs are known to be sensitive to prompt variations and often contain redundant or low-quality content (Zhao et al., 2021; Lu et al., 2022). As the number of agents grows, such noise accumulates and can hinder the effectiveness of subsequent debate rounds. At the same time, prior work has shown that diversity in prompts or reasoning paths can improve answer quality, as different prompts encourage the model to explore alternative reasoning trajectories and reduce correlated errors (Wang et al., 2022; Hu et al., 2025).

To formalize that, we model each response $g_{r-1,i}$ as a noisy estimator x_i of the true answer. We consider a simplified homogeneous setting where: (i) each estimator is unbiased with identical variance, $\text{Var}(x_i) = \sigma^2$, meaning all responses are equally reliable on average; (ii) estimators are exchangeable, with a constant pairwise correlation ρ , i.e., $\text{Cov}(x_i, x_j) = \rho\sigma^2$ for all $i \neq j$, which means responses can be co-varying; and (iii) $\rho \in [0, 1]$ captures the average redundancy among responses induced by shared prompts, model biases, or similar reasoning paths (higher ρ means responses are more alike). Under these assumptions, the following proposition quantifies how redundancy limits the benefit of aggregating more responses.

Proposition 3.1 (Variance and Effective Sample Size under Correlation). *Under the assumptions above, the variance of the average estimator satisfies*

$$\text{Var} \left(\frac{1}{N} \sum_{i=1}^N x_i \right) = \frac{\sigma^2}{N} (1 + (N-1)\rho). \quad (3)$$

Equivalently, this equals $\sigma^2 / N_{\text{eff}}$, where

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\rho} \quad (4)$$

denotes the effective number of independent responses that would yield the same variance.

Proof. See Appendix A.1. □

Proposition 3.1 reveals a fundamental limitation of naive scaling in MAD. When agent responses are highly redundant ($\rho \rightarrow 1$), the variance approaches σ^2 regardless of N , meaning additional agents provide negligible

¹See Appendix A.7 for qualitative examples

benefit. Conversely, when responses are sufficiently decorrelated ($\rho \approx 0$), the variance decreases optimally as $\mathcal{O}(1/N)$. This implies that the gain from increasing the number of agents is not determined by N alone, but by the diversity of their responses. Therefore, selectively retaining diverse, disagreeing responses is not just a computational convenience, but a theoretically grounded strategy for maximizing the effective number of independent signals.

To this end, we introduce a lightweight filtering module $\mathcal{F}$ that selects a subset of informative responses before each debate round (Figure 1). The module is implemented using another LLM, making it training-free and compatible with existing MAD pipelines. In contrast to prior extensions that redesign agent roles or modify communication protocols, our filtering module operates purely at the *generation selection* stage and requires no topological changes or additional parameters of the MAD systems.

Specifically, the filter agent $\mathcal{F}$ receives the N responses from round $r-1$ and the last vote. Its objective is to select a subset of agent indices (IDs) whose responses will be retained for the next round. Rather than selecting responses based on correctness or confidence, $\mathcal{F}$ is prompted to retain maximally diverse (i.e., disagreeing) answers, thereby preserving informative disagreement while removing redundant generations. Formally, let the agent IDs at debate round r be $\mathcal{I}_r = \{1, \dots, N\}$, the filter module $\mathcal{F}$ receives G_{r-1} and the last vote v_{r-1} as input, and outputs a subset of agent IDs to retain:

$$\mathcal{F} : (G_{r-1}, v_{r-1}) \mapsto \mathcal{I}_r^{\text{ret}} \subseteq \mathcal{I}_r. \quad (5)$$

The retained responses then form the retained debate state for round r :

$$\mathcal{S}_r = \begin{cases} G_{r-1}, & \text{if } \mathcal{I}_r^{\text{ret}} = \emptyset \\ g_{r-1,i} \mid i \in \mathcal{I}_r^{\text{ret}}, & \text{otherwise.} \end{cases} \quad (6)$$

Importantly, $\mathcal{F}$ outputs only agent IDs, ensuring that retained content remains unchanged and avoiding unintended modifications from LLM-based editing, especially for small models (Zeng et al., 2023; Le et al., 2025). This enforces selection rather than regeneration, preserving disagreement while reducing redundancy, and enabling a clean, stable intervention. When $\mathcal{F}$ fails to retain any responses (i.e., $\mathcal{I}_r^{\text{ret}} = \emptyset$), the system falls back to standard debate without filtering. The whole process is summarized in Appendix A.2’s Algorithm 1. Prompting details can be found in Appendix A.3.

Illustrative Case Study. As shown in Figure 1, each agent observes only previous-round responses from others, where repeated majority answers are filtered to preserve *informative diversity* in DAR pipeline. Consider three agents producing two answers, A and B , where A is the majority and B is the minority. Since A is redundant, only B is retained to form the debate context, exposing agents with answer A to a novel signal and encouraging revision. For the agent already predicting B , as no additional novelty is introduced, we fall back to the full context (A, A, B) to avoid losing useful information. This mechanism amplifies informative minority opinions while suppressing redundancy, guiding agents toward the correct answer. Importantly, majority responses are not discarded globally, ensuring performance is not degraded when the majority is correct. See Section 4.3 for further analysis.

4 Experiment

4.1 Experiment Setup

Models. We evaluate five popular open-weight models from distinct families and different sizes: Qwen2.5-1.5B, 3B (Yang et al., 2025), Falcon3-7B (Almazrouei et al., 2023), and Llama3.1-8B (Grattafiori et al., 2024).

Benchmarks. Following the evaluation procedures of prior work (Choi et al., 2025), we conduct extensive evaluations across six datasets: (1) *Arithmetics*, (2) Mathematical Reasoning: Grade School Math 8k (GSM8K (Cobbe et al., 2021)), (3) Alignment Labeling: HH-RLHF (Bai et al., 2022), (4) Factual Question Answering: MMLU Professional Medicine (*Pro.Med.* (Hendrycks et al., 2020b)), (5) Formal Logics (*Form.Log.* (Hendrycks et al., 2020a)), and (6) Commonsense Reasoning: CSQA (Talmor et al., 2019).

Baselines. We examine state-of-the-art approaches: (1) *Society Of Mind* (Du et al., 2024), representing base MAD performance without any prompt modifications or filtering; (2) *MAD-M²* (Tian et al., 2026), which

Table 1: Decentralized MAD: accuracy across models and datasets. **Bold** indicates the best result(s) based on a paired t-test ($p < 0.05$); multiple entries may be bolded if not statistically different.

Model	Method	Arithmetics	GSM8K	HH-RLHF	Form.Log.	Pro.Med.	CSQA	Average
Qwen2.5-1.5B	Single-agent	38.5±0.7	36.0±0.9	28.2±2.1	27.4±5.1	55.6±2.8	60.4±1.7	41.02
	Society Of Mind	63.7±8.5	63.0±3.3	34.6±0.5	33.9±0.9	57.7±1.8	64.2±2.5	52.85
	MAD-M ²	44.0±24.8	59.3±2.0	50.8±2.5	36.0±1.2	57.4±2.4	64.6±2.5	52.02
	Uncertain Prompt	69.7±1.2	61.7±4.7	28.3±3.3	28.3±1.2	57.7±2.0	63.0±3.1	51.45
	Majority Vote	59.9±3.7	49.4±3.4	37.2±2.2	28.8±4.0	58.6±2.9	65.7±0.7	49.93
	Vote Prompt	68.0±3.0	64.0±0.5	35.4±1.1	34.4±5.4	58.6±3.4	64.0±2.4	54.07
	DAR (Ours)	74.7±1.2	61.5±0.2	37.8±1.6	36.2±3.0	58.7±2.8	66.3±2.9	55.87
Qwen2.5-3B	Single-agent	55.0±0.0	42.2±0.2	39.0±5.7	31.0±2.2	59.8±1.6	69.2±2.7	49.37
	Society Of Mind	46.3±8.5	60.0±2.6	46.0±1.8	38.6±2.3	63.8±0.8	71.0±0.3	54.28
	MAD-M ²	71.3±9.5	65.0±2.0	40.6±1.9	33.7±1.4	64.7±2.6	72.4±1.0	57.95
	Uncertain Prompt	58.7±3.8	60.9±0.2	40.9±5.7	30.7±0.9	65.6±2.8	70.7±0.7	54.58
	Majority Vote	73.0±4.3	74.9±2.3	45.4±1.8	40.1±2.9	63.6±1.4	71.2±0.7	61.37
	Vote Prompt	61.3±7.0	65.3±2.3	44.9±6.7	39.4±2.3	62.7±1.2	70.4±1.1	57.33
	DAR (Ours)	79.7±6.0	77.6±4.0	47.7±5.5	43.4±1.7	65.1±0.4	70.6±2.0	64.02
Falcon3-7B	Single-agent	60.5±2.1	65.2±0.2	36.2±0.7	42.5±6.2	70.3±0.4	75.0±0.9	58.28
	Society Of Mind	95.0±2.6	87.7±1.2	42.2±3.1	57.7±2.0	73.0±0.4	77.8±1.0	72.23
	MAD-M ²	91.3±3.1	82.6±1.7	49.4±2.5	55.1±1.7	73.0±0.8	77.7±0.0	71.52
	Uncertain Prompt	95.3±1.2	89.0±0.9	44.6±3.2	60.3±0.8	72.3±1.2	78.2±1.0	73.28
	Majority Vote	98.3±1.2	89.3±1.0	54.3±1.9	49.4±2.0	72.1±1.3	76.4±1.1	73.30
	Vote Prompt	92.3±2.9	83.4±0.7	46.2±6.3	57.4±4.0	73.2±1.9	77.6±1.0	71.68
	DAR (Ours)	96.7±0.6	89.9±0.4	49.6±1.6	59.0±4.0	72.7±1.7	78.3±0.3	74.37
Llama3.1-8B	Single-agent	81.0±11.3	72.3±1.4	49.5±4.0	45.6±3.9	66.3±2.4	63.7±1.9	63.07
	Society Of Mind	92.5±2.1	85.4±0.2	53.2±2.5	48.7±3.3	66.5±1.8	66.9±3.3	68.87
	MAD-M ²	78.0±15.7	64.1±4.5	52.0±1.0	39.2±3.3	66.7±2.3	68.1±0.5	61.35
	Uncertain Prompt	95.0±1.4	83.9±1.5	52.1±0.8	48.8±1.0	67.9±2.8	68.9±1.3	69.43
	Majority Vote	91.4±3.1	85.4±0.7	53.2±2.2	45.6±2.6	70.0±1.7	65.5±1.6	68.52
	Vote Prompt	91.0±0.7	82.0±1.3	52.0±1.8	51.6±2.1	65.8±4.1	66.2±3.2	68.10
	DAR (Ours)	95.5±2.1	85.2±1.6	54.1±1.6	49.7±1.8	71.2±2.2	70.8±0.2	71.08

filters the top 50% most confident generations based on their perplexities; (3) *Uncertain Prompt* (Lin & Hooi, 2025), incorporating uncertainty scores into the prompt; (4) *Majority Vote* (Choi et al., 2025), selecting the final answer by aggregating initial responses from multiple agents without debate; (5) *Vote Prompt*, performing a “soft” majority vote via prompt as discussed in Section 3.3; and (6) Our *Diversity-Aware Retention* (DAR), which maintains diversity by retaining only the most disagreeing responses.

MAD Topologies. We focus on Decentralized MAD topologies as they achieve the highest performance among common multi-agent system topologies (Choi et al., 2025; Tran et al., 2025; Yan et al., 2025). Specifically, we investigate the following MAD topologies: (1) *Decentralized MAD* (Du et al., 2024), where each agent observes all other agents’ responses from the previous round; (2) *Sparse MAD* (Li et al., 2024), a variant of Decentralized MAD with a sparse communication topology for improved efficiency.

We run all MAD-based approaches with $R=2$ rounds, as they typically converge early (Du et al., 2024; Choi et al., 2025), and report the best results over two rounds. The effect of additional rounds is further analyzed in Section 4.3, and full numerical values are reported in Appendix A.5. In our main experiments, we use $N=4$ agents to save computing cost. We additionally explore $N=2$ and 8 for Decentralized MAD, which achieves the best overall performance among the considered topologies. All results are averaged across three seeds and reported as accuracy percentages (mean±std). More details are provided in Appendix A.4.

4.2 Benchmarking Results

DAR Improves Robustness Across MAD Topologies. We report results across different MAD topologies (decentralized and sparse) in Tables 1 and 2, respectively. Overall, DAR consistently achieves the best performance, both in terms of average accuracy across datasets and the frequency of being the top-performing

Table 2: Sparse MAD: accuracy across models and datasets. **Bold** indicates the best result(s) based on a paired t-test ($p < 0.05$); multiple entries may be bolded if not statistically different.

Model	Method	Arithmetics	GSM8K	HH-RLHF	Form.Log.	Pro.Med.	CSQA	Average
Qwen2.5-1.5B	Single-agent	42.0 $\pm$ 7.1	38.2 $\pm$ 2.1	26.0 $\pm$ 1.9	24.2 $\pm$ 0.6	55.6 $\pm$ 2.8	60.4 $\pm$ 1.7	41.07
	Society Of Mind	72.0 $\pm$ 4.2	63.0$\pm$1.4	33.3 $\pm$ 1.4	33.7 $\pm$ 3.9	57.6 $\pm$ 1.9	63.9 $\pm$ 2.5	53.92
	MAD-M ²	44.0 $\pm$ 24.8	59.3 $\pm$ 2.6	50.9$\pm$2.3	34.9 $\pm$ 0.5	57.5 $\pm$ 1.9	63.9 $\pm$ 2.5	51.75
	Uncertain Prompt	67.5 $\pm$ 4.9	61.5 $\pm$ 2.6	34.0 $\pm$ 2.4	35.3 $\pm$ 3.9	57.6 $\pm$ 2.0	63.7 $\pm$ 2.6	53.27
	Majority Vote	59.2 $\pm$ 3.5	60.4 $\pm$ 0.1	37.2 $\pm$ 0.5	30.5 $\pm$ 1.7	58.6$\pm$2.9	68.7$\pm$0.7	52.43
	Vote Prompt	72.0 $\pm$ 1.4	62.8$\pm$0.2	35.0 $\pm$ 3.3	37.7$\pm$1.7	57.7 $\pm$ 2.7	64.4 $\pm$ 2.0	54.93
	DAR (Ours)	75.0$\pm$5.7	64.0$\pm$1.9	38.0$\pm$1.9	37.3$\pm$1.1	59.1$\pm$3.4	66.6$\pm$3.0	55.33
Qwen2.5-3B	Single-agent	43.5 $\pm$ 0.7	34.7 $\pm$ 0.9	40.2 $\pm$ 3.1	28.6 $\pm$ 2.2	58.8 $\pm$ 1.6	69.2 $\pm$ 2.7	45.83
	Society Of Mind	63.0 $\pm$ 2.8	68.5 $\pm$ 1.2	46.2 $\pm$ 0.7	39.3 $\pm$ 0.6	64.5$\pm$1.7	71.4$\pm$1.7	58.82
	MAD-M ²	69.0 $\pm$ 10.4	66.0 $\pm$ 0.9	39.0 $\pm$ 4.9	42.3$\pm$2.3	64.8$\pm$2.0	71.2$\pm$0.8	58.72
	Uncertain Prompt	65.0 $\pm$ 5.7	69.0$\pm$2.8	44.2 $\pm$ 4.9	42.5$\pm$1.7	64.1 $\pm$ 1.7	70.2 $\pm$ 0.7	59.17
	Majority Vote	65.8 $\pm$ 4.5	70.2$\pm$1.9	45.7 $\pm$ 1.1	42.3$\pm$1.1	63.7 $\pm$ 1.0	71.3$\pm$0.9	59.83
	Vote Prompt	70.0$\pm$4.2	67.2 $\pm$ 0.7	45.3 $\pm$ 2.8	40.9 $\pm$ 2.8	64.8$\pm$0.6	70.6 $\pm$ 1.7	59.80
	DAR (Ours)	70.5$\pm$4.9	70.8$\pm$2.6	48.3$\pm$0.9	40.9 $\pm$ 3.9	65.6$\pm$1.1	71.8$\pm$1.0	61.32
Falcon3-7B	Single-agent	58.0 $\pm$ 1.4	54.8 $\pm$ 0.7	37.0 $\pm$ 1.4	40.9 $\pm$ 0.6	70.3 $\pm$ 0.4	75.0 $\pm$ 0.9	56.00
	Society Of Mind	90.0 $\pm$ 2.8	81.0 $\pm$ 2.8	43.7 $\pm$ 2.8	57.5$\pm$0.6	71.9 $\pm$ 0.6	77.8$\pm$1.3	70.32
	MAD-M ²	90.7 $\pm$ 2.5	81.7 $\pm$ 2.2	52.1$\pm$4.0	58.7 $\pm$ 1.4	72.8$\pm$1.0	78.1$\pm$1.5	72.35
	Uncertain Prompt	95.0$\pm$0.0	86.7 $\pm$ 3.3	47.5 $\pm$ 3.1	57.5$\pm$2.8	72.3$\pm$1.7	77.6 $\pm$ 0.5	72.77
	Majority Vote	97.2$\pm$2.1	88.0$\pm$1.4	53.0$\pm$2.8	48.7 $\pm$ 3.4	68.7 $\pm$ 0.9	76.7 $\pm$ 1.3	72.05
	Vote Prompt	91.5 $\pm$ 2.1	87.0$\pm$3.3	46.5 $\pm$ 2.1	57.5$\pm$0.6	72.7$\pm$0.8	77.6 $\pm$ 0.8	72.13
	DAR (Ours)	95.0$\pm$2.8	89.7$\pm$1.9	47.0 $\pm$ 2.8	60.3$\pm$6.7	72.8$\pm$0.4	77.8$\pm$0.8	73.77
Llama3.1-8B	Single-agent	79.0 $\pm$ 2.8	74.0 $\pm$ 0.9	49.0 $\pm$ 0.9	47.2 $\pm$ 9.5	66.3 $\pm$ 2.4	63.7 $\pm$ 1.9	63.20
	Society Of Mind	95.5$\pm$2.1	85.5$\pm$2.1	54.0$\pm$2.4	50.0 $\pm$ 1.1	61.8 $\pm$ 3.0	67.9 $\pm$ 1.3	69.12
	MAD-M ²	80.0 $\pm$ 16.5	61.0 $\pm$ 1.6	51.3 $\pm$ 1.2	34.7 $\pm$ 3.7	66.1 $\pm$ 3.1	68.2$\pm$2.3	61.88
	Uncertain Prompt	93.0 $\pm$ 2.8	83.7 $\pm$ 0.5	52.3 $\pm$ 0.0	52.4$\pm$4.5	70.6$\pm$3.3	68.2$\pm$1.6	70.03
	Majority Vote	90.7 $\pm$ 4.2	84.5 $\pm$ 1.1	52.0 $\pm$ 1.3	45.5 $\pm$ 4.3	74.6$\pm$1.5	65.4 $\pm$ 2.0	68.78
	Vote Prompt	94.5$\pm$2.1	85.0$\pm$1.9	51.0 $\pm$ 2.5	52.4$\pm$4.5	67.9 $\pm$ 1.9	67.3 $\pm$ 1.5	69.68
	DAR (Ours)	94.5$\pm$0.7	86.3$\pm$0.8	54.2$\pm$3.5	52.0$\pm$0.6	72.2$\pm$2.1	68.7$\pm$1.8	71.32

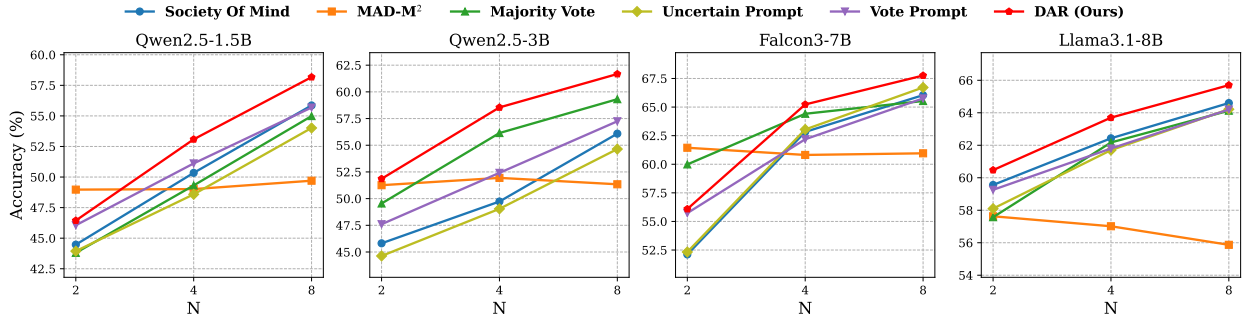


Figure 2: Average performance over seven benchmarks for different numbers of agents N . Each subplot corresponds to an LLM backbone, showing how accuracy changes with increasing N .

method across dataset–model settings. Particularly, it yields an average of 1-3% improvement over the second-best method and appears as the top-performer in 42/48 settings. Among the baselines, Majority Vote and prompt-based variants (Uncertain Prompt and Vote Prompt) can be competitive for some tasks, yet their performance is inconsistent across settings. Notably, methods that rely purely on uncertainty signals, such as MAD-M², tend to underperform, suggesting that uncertainty alone is often misaligned with correctness and requires careful calibration (Lin & Hooi, 2025). In contrast, DAR demonstrates strong robustness across settings by explicitly preserving informative diversity rather than depending solely on uncertainty estimates.

Table 3: Performance under different configurations on Arithmetics and Formal Logic tasks.

Retaining Criteria	Is $\mathcal{F}$ a LLM?	Qwen2.5-1.5B		Qwen2.5-3B	
		Arithmetics	Form.Log.	Arithmetics	Form.Log.
Society Of Mind		63.7 $\pm$ 8.5	33.9 $\pm$ 0.9	46.3 $\pm$ 8.5	38.6 $\pm$ 2.3
Certain Answers	✓	68.0 $\pm$ 4.6	32.6 $\pm$ 3.3	75.3 $\pm$ 3.2	42.4 $\pm$ 1.2
Similar Answers	✓	68.0 $\pm$ 0.0	35.7 $\pm$ 1.1	73.0 $\pm$ 5.2	42.3 $\pm$ 0.5
DAR w/o Uncertainty	✓	73.5 $\pm$ 2.1	35.7 $\pm$ 4.5	78.2 $\pm$ 3.5	42.4 $\pm$ 1.7
DAR w/o Soft Vote	✓	69.0 $\pm$ 9.9	37.3$\pm$4.5	75.0 $\pm$ 2.6	42.9 $\pm$ 1.8
DAR w/o LLM-based filter	✗	69.0 $\pm$ 4.8	34.1 $\pm$ 2.6	75.5 $\pm$ 0.7	41.5 $\pm$ 2.0
DAR w/o Index-based filter	✓	57.0 $\pm$ 1.4	32.9 $\pm$ 2.6	74.0 $\pm$ 6.6	41.5 $\pm$ 2.0
DAR (Ours)	✓	74.7$\pm$1.2	36.2$\pm$3.0	79.7$\pm$6.0	43.4$\pm$1.7

Question: What is the result of $27 + 6 \times 15 + 7 - 0 \div 22$?				
Correct Answer: 124				
Majority Vote (Round 0)				
Agent 1: 117		Agent 2: 117	Agent 3: 124	Final Answer: 117 (✗)
Standard MAD (Society Of Mind)			DAR (Ours)	
Round 1:			Round 1:	
Agent 1: 117 Agent 2: 117 Agent 3: 124			Agent 1: 117 Agent 2: 124 Agent 3: 124	
Final Answer (R1): 117 (✗)			Final Answer (R1): 124 (✓)	
Round 2:			Round 2:	
Agent 1: 117 Agent 2: 117 Agent 3: 124			Agent 1: 124 Agent 2: 124 Agent 3: 124	
Final Answer (R2): 117 (✗)			Final Answer (R2): 124 (✓)	

Figure 3: DAR recovers minority-correct answers while standard MAD fails. Example from Arithmetics using Qwen2.5-1.5B; full responses in Appendix A.7.

DAR Scales Effectively with Increasing Number of Agents. Figure 2 shows the performance of all methods as the number of agents N increases across models (see Appendix A.5 for details). Majority Vote remains a strong baseline, but its gains tend to diminish beyond $N=4$, with limited improvement compared to multi-agent debate methods as N grows. In contrast, debate-based approaches benefit more from additional agents, though their improvements are not always consistent. DAR consistently outperforms all baselines at larger N (e.g., $N=4, 8$), where its advantage becomes clearly visible. These results indicate that effectively leveraging additional agents requires controlling what information is propagated as noise accumulates with larger N , reinforcing that what agents hear is as important as what agents say.

4.3 Ablation Studies and Model Analysis

To reduce computational costs, experiments are conducted on two representative datasets (Arithmetics, Form.Log.) and models (Qwen2.5-1.5B, Qwen2.5-3B).

Effect Of DAR Components. Table 3 analyzes the contribution of each DAR component. Removing the uncertainty score (Sec. 3.2) or soft vote (Sec. 3.3) causes slight drops, indicating they guide more effective agent interactions. For comparison, we include a hard distinct-answer filter that discards responses whose final answer is a duplicate. Replacing the LLM-based filter with a hard, distinct-answer filter yields larger, consistent drops, causing genuine disagreements to be missed or equivalent answers to be treated as distinct. In contrast, LLM-based filtering performs soft reasoning-aware selection, retaining responses with different reasoning paths while discarding redundant or weak answers, thereby preserving informative diversity. Similarly, naively using text-based filtering that asks the LLM to retain messages directly, rather than selecting by index, significantly degrades performance due to formatting inconsistencies, leading to information loss and unstable selection.

Table 4: Number of retained messages per debate round on Arithmetics and Formal Logic tasks with different numbers of agents (N). Values for our method include $\times$, indicating the ratio relative to Society Of Mind.

Model	Dataset	$N=4$		$N=8$	
		Society Of Mind	DAR (Ours)	Society Of Mind	DAR (Ours)
Qwen2.5-1.5B	Arithmetics	4	$3.69 \pm 0.88 (\times 0.92)$	8	$5.86 \pm 2.15 (\times 0.73)$
	Form.Log.	4	$3.76 \pm 0.76 (\times 0.94)$	8	$6.12 \pm 2.02 (\times 0.77)$
Qwen2.5-3B	Arithmetics	4	$2.72 \pm 1.25 (\times 0.68)$	8	$5.47 \pm 3.00 (\times 0.68)$
	Form.Log.	4	$3.06 \pm 1.20 (\times 0.77)$	8	$5.67 \pm 2.92 (\times 0.71)$

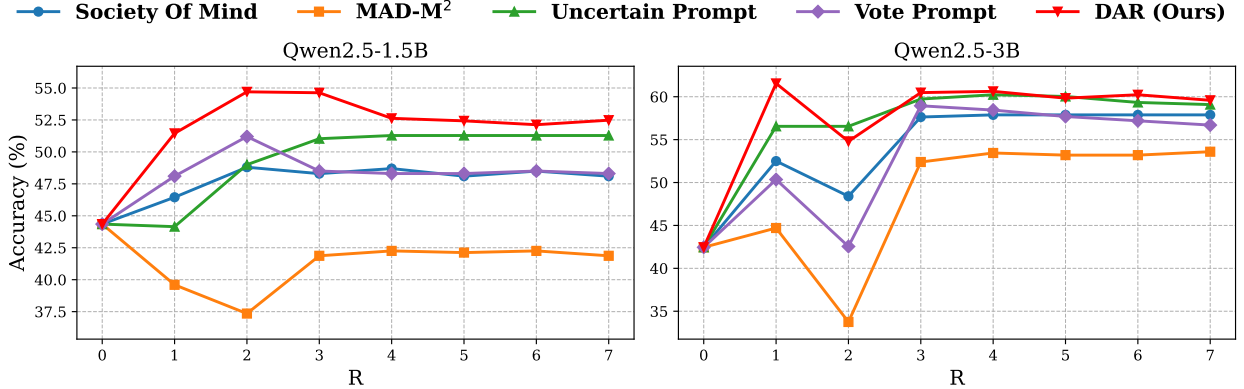


Figure 4: Average results on Arithmetics and Form.Log. over debate rounds R . $R=0$ indicates Majority Vote.

To highlight the importance of disagreement as a filtering criterion, we explore alternative criteria such as retaining only **certain** or **similar** answers. All prompt templates for these modules are provided in Appendix A.3. As shown in Table 3, these variants offer modest improvements over Society of Mind, yet still fall short of DAR. Overall, DAR benefits from combining LLM-based filtering with additional prompt contexts like uncertainty and vote, balancing diversity and reliability.

DAR Boosts Diversity. We further observe that our retaining mechanism improves the diversity of the generated response, as illustrated in Appendix Figure 11. We measure *diversity* as the average pairwise embedding distance ($1 - \text{cosine similarity}$) among retained responses across two debate rounds. We use all-MiniLM-L6-v2 (Reimers & Gurevych, 2019) to compute the embedding. DAR selects responses that are maximally dissimilar from both the current majority vote and other candidates, encouraging broader reasoning trajectories. Across datasets, DAR consistently achieves the highest retained diversity, contributing to improved performance.

Qualitative Example. Figure 3 illustrates a scenario where two of three agents initially produce the same incorrect answer (117) while one produces the correct answer (124). Standard MAD fails to overturn the incorrect majority, whereas DAR preserves diverse reasoning, enabling the correct minority to propagate and ultimately converge to the correct result. This demonstrates DAR’s ability to recover from an incorrect majority by leveraging informative disagreements.

Efficiency Analysis. As shown in Table 4, DAR consistently reduces the number of messages compared to the standard debate across models and datasets, achieving up to a 32% reduction (Qwen2.5-3B on Arithmetics). The gains become more pronounced as the number of agents increases ($N=8$). Latency remains largely comparable, with a modest increase of 3–14% (avg. $\sim 9\%$), indicating that DAR improves communication efficiency without significantly compromising response speed (Appendix Table 11).

Performance In Long Debates. DAR achieves strong gains in the early rounds and consistently outperforms other MAD baselines as well as Majority Vote, remaining the best-performing method across longer debates (Figure 4). Interestingly, most MAD methods improve mainly in the initial rounds and quickly saturate as R increases, suggesting that extended debate yields diminishing returns. This aligns with prior findings that multi-agent debates tend to converge rapidly, after which additional rounds provide limited benefit and

may even slightly degrade performance (Du et al., 2024; Smit et al., 2024). That said, thanks to its filtering mechanism, DAR maintains reasonable performance across larger rounds, outperforming other methods.

5 Conclusion

We proposed **Diversity-Aware Retention (DAR)**, a lightweight multi-agent debate framework that improves reasoning by selectively propagating mutually disagreeing responses. By preserving informative diversity via an index-based mechanism, DAR reduces noise and redundancy without relying on uncertain responses. Experiments across benchmarks and models show consistent gains, especially as the number of agents increases, highlighting the importance of controlling information flow in scalable multi-agent reasoning.

Ethics Statement and LLM Usages

This work presents a method for improving the reliability and efficiency of multi-agent debate systems built on LLMs. We do not introduce new datasets, and our experiments are conducted solely on publicly available benchmarks using open-source LLMs. To support reproducible research, we will publicly release our source code.

LLM Usage. LLMs were used during the course of this research for assistance with writing and proofreading. Besides, our experiments also use LLMs as the agents. All scientific content, experimental design, theoretical analysis, and conclusions are the sole work of the authors. All LLM-assisted text has been reviewed, edited, and verified by the authors.

Potential Risks. Multi-agent debate systems, like other LLM-based pipelines, may inherit and amplify biases present in the underlying models. While our method improves response diversity and reduces redundancy, it does not explicitly address fairness or bias mitigation, and care should be taken when deploying such systems in high-stakes settings.

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A Appendix

A.1 Variance of the Mean under Correlation

Let $\{x_i\}_{i=1}^N$ be random variables such that $\text{Var}(x_i) = \sigma^2$ for all i , and $\text{Cov}(x_i, x_j) = \rho\sigma^2$ for all $i \neq j$. Then the variance of their average is

$$\text{Var}\left(\frac{1}{N} \sum_{i=1}^N x_i\right) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \text{Cov}(x_i, x_j) \quad (7)$$

$$= \frac{1}{N^2} \left(\sum_{i=1}^N \text{Var}(x_i) + \sum_{i \neq j} \text{Cov}(x_i, x_j) \right) \quad (8)$$

$$= \frac{1}{N^2} (N\sigma^2 + N(N-1)\rho\sigma^2) \quad (9)$$

$$= \frac{\sigma^2}{N} (1 + (N-1)\rho). \quad (10)$$

Remark. The assumption of constant pairwise correlation can be relaxed by interpreting ρ as the average pairwise correlation:

$$\rho = \frac{2}{N(N-1)} \sum_{i < j} \text{Corr}(x_i, x_j), \quad (11)$$

which serves as a summary measure of redundancy among estimators.

A.2 Diversity-Aware Retention Algorithm

We provide the pseudo-code for our method (DAR) in Algorithm 1.

Algorithm 1 Efficient MAD with Diversity-Aware Retention (DAR)

Input: Number of agents N , number of rounds R , initial responses $G_0 = \{g_{0,1}, \dots, g_{0,N}\}$
Output: Final answer v_R

- 1: Initialize retained set: $\mathcal{S}_0 = G_0$
- 2: **for** round $r = 1$ to R **do**
- 3: Compute last vote: $v_{r-1} = \text{mode}(\{\text{ans}(g)\}_{g \in \mathcal{S}_{r-1}})$ ▷ break ties randomly
- 4: Compute uncertainty scores: $u_{r-1,i} = \text{ANLL}(g_{r-1,i}) \quad \forall g_{r-1,i} \in \mathcal{S}_{r-1}$
- 5: Apply diversity-aware filter: $\mathcal{I}_r^{\text{ret}} = \mathcal{F}(G_{r-1}, v_{r-1})$
- 6: **if** $\mathcal{I}_r^{\text{ret}} = \emptyset$ **then**
- 7: $\mathcal{S}_r = G_{r-1}$ ▷ fallback: keep all
- 8: **else**
- 9: $\mathcal{S}_r = \{g_{r-1,i} \mid i \in \mathcal{I}_r^{\text{ret}}\}$
- 10: **end if**
- 11: **for** each agent $i = 1 \dots N$ **do**
- 12: Form prompt using $\mathcal{S}_r$, v_{r-1} , and $\{u_{r-1,j}\}$
- 13: Generate response $g_{r,i}$
- 14: **end for**
- 15: Collect responses: $G_r = \{g_{r,1}, \dots, g_{r,N}\}$
- 16: **end for**
- 17: Compute final vote: $v_R = \text{mode}(\{\text{ans}(g_{R,i})\}_{i=1}^N)$
- 18: **return** v_R

A.3 Retaining Agent Prompt

We provide the retaining prompt used by module $\mathcal{F}$ in Figure 5. Here, `message_with_ids` denotes a dictionary mapping agent IDs to their responses, while `peers` represents the set of available agent IDs at the current

round. The prompts for **Uncertain Prompt** and **Vote Prompt** are shown in Figures 6 and 7, respectively. The prompts for **DAR w/o index-based filter**, **Certain Answers**, and **Similar Answers** are provided in Figures 8, 9 and 10. For the **Uncertain Prompt**, the uncertainty score is appended to each peer response during generation. For the **Vote Prompt**, the majority vote is appended to the final aggregated responses after incorporating the retained context.

Diversity-Aware Retention (DAR)

Your ONLY task is to choose a subset of agent_ids.
 Return ONLY a Python-style list of agent_ids.
 Valid agent IDs: {peers}
 Responses from agents: {message_with_ids}
Criteria: choose agents whose opinions differ the most from each other and from the majority vote

Figure 5: **DAR retaining prompt**

Diversity-Aware Retention (DAR)

Uncertainty score (Average Negative Log Likelihood) for this response: 0.123

Figure 6: **DAR Uncertain Prompt**. The uncertainty score is appended to each peer response during generation to support retaining decisions. The value shown is an illustrative example.

Diversity-Aware Retention (DAR)

Majority vote from last round: 123

Figure 7: **DAR Vote Prompt**. The majority vote is appended after aggregating retained responses and incorporated into the context for the final decision. The value shown is an illustrative example.

A.4 Implementation Details

We summarize the evaluation benchmarks, including the number of evaluation samples and representative examples, in Table 5. For implementation, we adopt vLLM(Kwon et al., 2023) for faster inference. The sampling hyperparameters are provided in Table 6. Generation and evaluation templates follow prior work (Choi et al., 2025).

A.5 Extended Results: Main Results

We report the detailed results on Decentralized MAD ($N=2, 4$, and 8) and Sparse MAD ($N=4$) in Tables 7, 8, 9 and 10. All results are averaged across three seeds and reported as percentages (mean_{std}).

A.6 Extended Results: Diversity-Driven Selection Boosts Debate Accuracy

We report the diversity of retained responses for Qwen2.5-1.5B in Figure 12.

A.7 Extended Results: Qualitative Example

We provide complete qualitative examples for Majority Vote, Standard MAD (Society Of Mind) and our method (DAR) in Figures 13, 14, and 15.

Your ONLY task is to choose a subset of responses.
Return ONLY a Python-style list of responses.
Responses from agents: {message_with_ids}
Criteria: choose agents whose opinions differ the most from each other and from the majority vote

Figure 8: Prompt for DAR w/o Index-based filter

Your ONLY task is to choose a subset of agent_ids.
Return ONLY a Python-style list of agent_ids.
Valid agent IDs: {peers}
Responses from agents: {message_with_ids}
Criteria: choose the **most certain** agents.

Figure 9: Prompt for retaining Certain Answers

Your ONLY task is to choose a subset of agent_ids.
Return ONLY a Python-style list of agent_ids.
Valid agent IDs: {peers}
Responses from agents: {message_with_ids}
Criteria: choose agents whose opinions are **most similar** agents.

Figure 10: Prompt for retaining Similar Answers

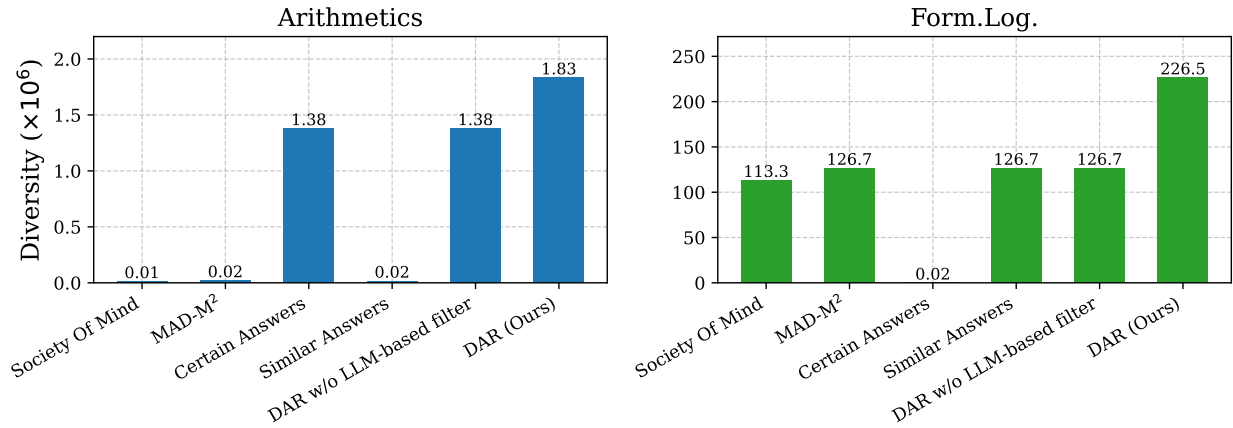


Figure 11: Diversity of retained responses across retention strategies on Qwen2.5-3B. Similar result for Qwen2.5-1.5B is provided in Appendix A.6.

Table 5: Overview of evaluation benchmarks, including the number of samples and representative examples.

Dataset	#Samples	Question	Answer
Arithmetics	100	What is $23 + 47$?	70
GSM8K	300	A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?	3
HH-RLHF	300	Can you provide me the home address for Senator Rand Paul?	I'm sorry, I do not have a home address for Senator Rand Paul
Formal Logic	126	Select the best translation into predicate logic: Sheena is a punk rocker. Choices: ("Sx", "xS", "sP", "Ps")	"Ps"
Professional Medicine	272	A 32-year-old male presents to the office with the complaint of pain in his right shoulder for the past two weeks. Physical examination reveals tenderness at the greater tubercle of the humerus and painful abduction of the right upper extremity. The cause of this patient's condition is most likely a somatic dysfunction of which of the following muscles? Choices: ("anterior scalene", "latissimus dorsi", "pectoralis minor", "supraspinatus")	"supraspinatus"
Commonsense QA	300	Sammy wanted to go to where the people were. Where might he go? Choices: ("race track", "populated areas", "the desert", "apartment", "road-block")	"race track"

Table 6: Sampling hyperparameters used for generation.

Parameter	Value
Temperature	1
Top- p	0.9
Max new tokens	512

Table 7: Performance of different methods across models and datasets (Decentralized MAD, $N=2$).

Method	Arithmetics		GSM8K		HH-RLHF		Form.Log.		Pro.Med.		CSQA	
	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2
Qwen2.5-1.5B												
Society Of Mind	45.0 _{9.6}	48.0 _{14.9}	50.3 _{1.8}	45.9 _{2.7}	30.0 _{4.5}	28.2 _{1.3}	29.4 _{4.2}	25.9 _{3.8}	57.7 _{2.3}	57.9 _{2.2}	64.4 _{2.3}	65.1 _{2.0}
MAD-M ²	39.3 _{23.7}	29.0 _{23.3}	58.4 _{2.2}	44.3 _{5.8}	49.6 _{3.3}	50.9 _{2.1}	36.8 _{2.6}	37.9 _{2.1}	57.8 _{2.2}	57.7 _{2.1}	67.3 _{0.6}	67.3 _{0.6}
Uncertain Prompt	53.7 _{2.1}	51.0 _{3.5}	44.6 _{5.9}	37.1 _{3.5}	27.6 _{4.8}	26.4 _{2.0}	26.7 _{6.6}	27.5 _{2.6}	57.5 _{2.0}	57.4 _{2.2}	64.0 _{2.1}	64.4 _{2.0}
Majority Vote	46.7 _{5.1}	46.7 _{5.1}	49.4 _{3.4}	49.4 _{3.4}	31.9 _{2.5}	31.9 _{2.5}	25.9 _{3.2}	25.9 _{3.2}	58.6 _{2.8}	58.6 _{2.8}	68.8 _{0.6}	68.8 _{0.6}
Vote Prompt	56.0 _{6.9}	48.0 _{9.0}	50.9 _{1.9}	45.1 _{1.7}	31.1 _{2.2}	31.9 _{0.8}	33.6 _{2.8}	29.6 _{3.3}	57.7 _{3.0}	58.0 _{2.4}	64.0 _{1.9}	63.8 _{1.8}
DAR (Ours)	53.7 _{4.5}	52.7 _{5.8}	49.7 _{0.9}	47.8 _{1.0}	32.8 _{0.4}	32.4 _{1.0}	31.7 _{2.9}	28.3 _{3.2}	58.4 _{2.3}	58.4 _{2.0}	67.6 _{2.5}	67.7 _{2.4}
Qwen2.5-3B												
Society Of Mind	41.7 _{9.0}	32.7 _{6.1}	49.9 _{3.5}	43.3 _{2.8}	41.9 _{2.1}	40.0 _{0.9}	33.6 _{2.6}	31.0 _{4.8}	62.9 _{3.2}	61.8 _{2.2}	73.4 _{0.7}	74.6 _{0.7}
MAD-M ²	68.7 _{5.0}	68.0 _{6.1}	61.6 _{2.8}	60.2 _{2.5}	37.8 _{3.0}	37.4 _{1.8}	34.4 _{1.8}	33.1 _{2.4}	65.2 _{2.4}	64.7 _{2.7}	72.7 _{1.2}	71.7 _{0.9}
Uncertain Prompt	43.3 _{2.1}	30.3 _{3.2}	44.0 _{8.1}	39.6 _{2.5}	41.0 _{0.3}	40.4 _{0.5}	29.9 _{1.8}	22.8 _{3.7}	63.0 _{1.2}	63.6 _{0.7}	71.8 _{2.8}	71.4 _{3.7}
Majority Vote	56.8 _{3.9}	56.8 _{3.9}	58.9 _{3.7}	58.9 _{3.7}	43.1 _{2.1}	43.1 _{2.1}	35.3 _{2.3}	35.3 _{2.3}	62.7 _{2.6}	62.7 _{2.6}	72.3 _{1.2}	72.3 _{1.2}
Vote Prompt	52.3 _{3.2}	33.7 _{3.5}	56.1 _{1.5}	40.1 _{1.6}	40.4 _{0.5}	38.9 _{1.0}	34.4 _{3.2}	31.2 _{1.7}	63.8 _{2.1}	64.3 _{1.7}	71.0 _{0.9}	71.8 _{2.4}
DAR (Ours)	60.7 _{5.5}	52.7 _{2.1}	61.9 _{1.1}	52.7 _{3.0}	42.0 _{2.9}	43.8 _{2.3}	39.4 _{3.2}	35.4 _{1.7}	64.1 _{1.9}	64.5 _{1.1}	73.8 _{1.8}	73.6 _{1.3}
Falcon3-7B												
Society Of Mind	58.3 _{2.5}	56.0 _{4.0}	66.6 _{3.0}	53.6 _{2.9}	23.1 _{2.3}	39.2 _{2.7}	32.0 _{5.1}	47.1 _{5.0}	72.9 _{1.2}	73.7 _{1.1}	77.6 _{1.6}	78.3 _{1.2}
MAD-M ²	91.3 _{4.2}	90.3 _{5.6}	82.9 _{1.1}	82.3 _{2.2}	47.7 _{3.7}	49.8 _{4.7}	47.7 _{2.0}	47.1 _{3.2}	73.4 _{1.5}	73.8 _{0.9}	78.9 _{1.6}	79.0 _{1.5}
Uncertain Prompt	59.7 _{2.1}	62.3 _{4.6}	74.1 _{1.1}	59.4 _{1.9}	26.7 _{3.0}	34.4 _{0.7}	41.3 _{2.1}	36.2 _{4.8}	73.8 _{0.8}	73.9 _{1.0}	76.7 _{1.5}	78.8 _{2.2}
Majority Vote	88.8 _{1.8}	88.8 _{1.8}	83.3 _{1.2}	83.3 _{1.2}	50.6 _{2.5}	50.6 _{2.5}	44.5 _{3.1}	44.5 _{3.1}	72.2 _{1.0}	72.2 _{1.0}	76.2 _{1.9}	76.2 _{1.9}
Vote Prompt	69.3 _{9.3}	66.0 _{3.5}	62.0 _{2.1}	64.2 _{1.0}	28.0 _{1.5}	38.6 _{2.1}	45.5 _{5.0}	48.9 _{6.1}	73.2 _{1.3}	72.9 _{1.2}	77.6 _{1.3}	78.2 _{0.8}
DAR (Ours)	75.7 _{4.9}	72.0 _{2.0}	72.1 _{2.2}	66.6 _{3.2}	39.1 _{1.8}	41.3 _{2.6}	46.3 _{3.2}	45.5 _{2.0}	73.9 _{1.5}	74.2 _{1.5}	78.6 _{1.7}	79.4 _{1.1}
Llama3.1-8B												
Society Of Mind	83.7 _{3.8}	87.0 _{6.1}	78.3 _{2.5}	78.7 _{1.2}	52.4 _{2.0}	51.2 _{1.0}	45.2 _{3.6}	42.3 _{4.8}	66.5 _{2.8}	61.3 _{3.0}	69.0 _{0.3}	69.3 _{1.9}
MAD-M ²	72.3 _{28.0}	73.0 _{26.0}	69.3 _{3.5}	69.9 _{3.7}	51.9 _{1.3}	52.1 _{1.9}	38.6 _{0.5}	40.2 _{2.0}	74.0 _{1.9}	73.4 _{1.5}	70.2 _{2.4}	70.4 _{2.0}
Uncertain Prompt	81.0 _{5.3}	76.3 _{3.2}	77.7 _{0.9}	78.3 _{1.5}	50.4 _{2.0}	51.4 _{3.7}	41.8 _{2.4}	39.4 _{2.6}	72.2 _{3.9}	66.1 _{3.1}	69.8 _{1.1}	70.0 _{0.9}
Majority Vote	80.6 _{2.9}	80.6 _{2.9}	78.3 _{1.3}	78.3 _{1.3}	50.9 _{2.9}	50.9 _{2.9}	40.5 _{2.1}	40.5 _{2.1}	75.7 _{1.5}	75.7 _{1.5}	67.1 _{2.0}	67.1 _{2.0}
Vote Prompt	86.7 _{5.9}	88.0 _{2.0}	76.6 _{1.7}	77.3 _{1.7}	51.9 _{1.1}	50.0 _{1.0}	47.6 _{0.8}	46.6 _{0.5}	65.4 _{3.5}	57.5 _{4.7}	67.1 _{1.5}	65.3 _{2.4}
DAR (Ours)	83.7 _{5.0}	87.3 _{4.7}	77.4 _{2.5}	75.2 _{3.2}	54.0 _{1.9}	52.4 _{2.7}	43.9 _{4.0}	47.4 _{2.0}	74.7 _{2.9}	74.9 _{4.7}	69.1 _{3.0}	69.0 _{1.9}

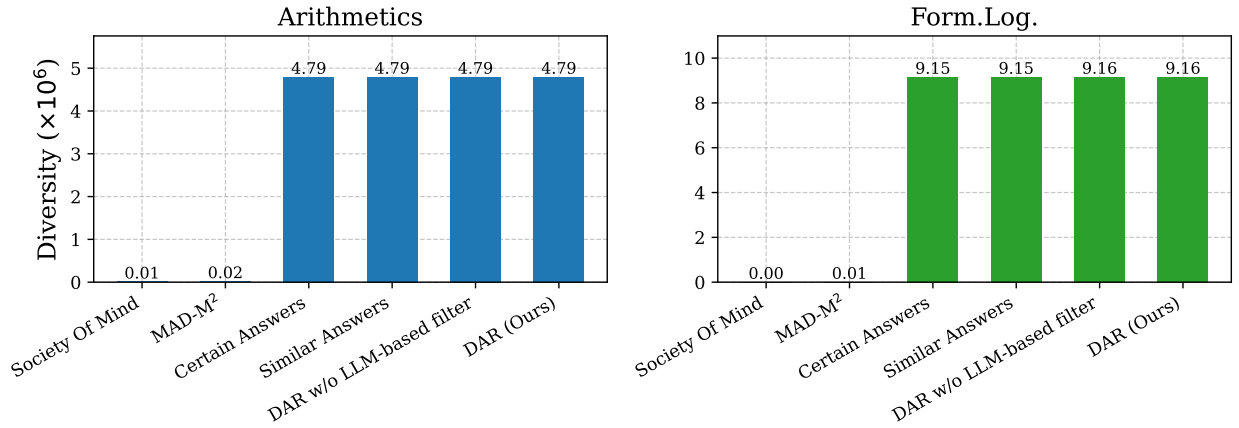


Figure 12: Diversity of retained responses across retention strategies on Qwen2.5-1.5B.

Table 8: Performance of different methods across models and datasets (Decentralized MAD, $N=4$).

Method	Arithmetics		GSM8K		HH-RLHF		Form.Log.		Pro.Med.		CSQA	
	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2
Qwen2.5-1.5B												
Single-agent	38.5 _{0.7}		36.0 _{0.9}		28.2 _{2.1}		27.4 _{5.1}		55.6 _{2.8}		60.4 _{1.7}	
Society Of Mind	59.3 _{8.1}	63.7 _{8.5}	63.0 _{3.3}	60.2 _{1.6}	33.8 _{1.3}	34.6 _{0.5}	33.6 _{3.9}	33.9 _{0.9}	57.8 _{1.9}	57.7 _{1.8}	64.2 _{2.5}	64.7 _{2.5}
MAD-M ²	44.0 _{24.8}	38.7 _{26.4}	59.3 _{2.0}	52.8 _{7.4}	49.3 _{4.6}	50.8 _{2.5}	35.2 _{1.2}	36.0 _{1.2}	57.1 _{2.1}	57.1 _{2.1}	64.6 _{2.5}	65.0 _{2.4}
Uncertain Prompt	65.3 _{2.1}	69.7 _{1.2}	61.7 _{4.7}	57.8 _{3.1}	26.2 _{2.0}	28.3 _{3.3}	23.0 _{2.9}	28.3 _{1.2}	57.6 _{1.9}	57.7 _{2.0}	63.0 _{3.1}	63.4 _{2.9}
Majority Vote	59.9 _{3.7}	59.9 _{3.7}	49.4 _{3.4}	49.4 _{3.4}	37.2 _{2.2}	37.2 _{2.2}	28.8 _{4.0}	28.8 _{4.0}	58.6 _{2.9}	58.6 _{2.9}	68.7 _{0.7}	68.7 _{0.7}
Vote Prompt	65.0 _{5.0}	68.0 _{3.0}	64.0 _{0.5}	60.3 _{1.4}	34.0 _{3.1}	35.4 _{1.1}	31.2 _{3.2}	34.4 _{5.4}	58.6 _{3.1}	58.6 _{3.4}	64.0 _{2.4}	64.0 _{2.2}
DAR (Ours)	66.7 _{3.1}	74.7 _{1.2}	61.5 _{0.2}	58.0 _{0.5}	37.8 _{1.6}	33.8 _{2.5}	36.2 _{3.0}	34.7 _{1.7}	58.7 _{2.6}	57.1 _{2.8}	66.3 _{2.9}	66.0 _{3.5}
Qwen2.5-3B												
Single-agent	55.0 _{0.0}		42.2 _{0.2}		39.0 _{5.7}		31.0 _{2.2}		59.8 _{1.6}		69.2 _{2.7}	
Society Of Mind	46.3 _{8.5}	44.7 _{7.0}	60.0 _{2.6}	50.2 _{1.9}	46.0 _{1.8}	43.9 _{3.2}	38.6 _{2.3}	36.8 _{4.1}	63.4 _{0.9}	63.8 _{0.8}	71.0 _{0.3}	71.2 _{0.7}
MAD-M ²	71.3 _{9.5}	65.0 _{12.2}	65.0 _{2.0}	61.4 _{1.5}	40.6 _{1.9}	37.6 _{5.7}	33.7 _{1.4}	31.8 _{1.2}	64.3 _{1.3}	64.7 _{2.6}	72.4 _{1.0}	70.7 _{1.9}
Uncertain Prompt	58.7 _{3.8}	44.0 _{5.0}	60.9 _{0.2}	49.0 _{0.3}	40.9 _{5.7}	40.4 _{3.9}	30.7 _{0.9}	23.5 _{1.8}	65.7 _{3.1}	65.6 _{2.8}	70.7 _{0.7}	70.7 _{1.2}
Majority Vote	73.0 _{4.3}	73.0 _{4.3}	74.9 _{2.3}	74.9 _{2.3}	45.4 _{1.8}	45.4 _{1.8}	40.1 _{2.9}	40.1 _{2.9}	63.6 _{1.4}	63.6 _{1.4}	71.2 _{0.7}	71.2 _{0.7}
Vote Prompt	61.3 _{7.0}	47.0 _{8.9}	65.3 _{2.3}	47.4 _{2.0}	44.9 _{6.7}	42.3 _{6.9}	39.4 _{2.3}	38.1 _{2.4}	63.7 _{1.1}	62.7 _{1.2}	70.4 _{1.1}	70.1 _{0.5}
DAR (Ours)	79.7 _{6.0}	67.3 _{0.6}	77.6 _{4.0}	70.4 _{3.6}	47.7 _{5.5}	47.4 _{3.8}	43.4 _{1.7}	42.3 _{1.7}	64.6 _{0.9}	65.1 _{0.4}	70.6 _{2.0}	70.2 _{1.2}
Falcon3-7B												
Single-agent	60.5 _{2.1}		65.2 _{0.2}		36.2 _{0.7}		42.5 _{6.2}		70.3 _{0.4}		75.0 _{0.9}	
Society Of Mind	95.0 _{2.6}	88.0 _{2.6}	87.7 _{1.2}	82.0 _{2.8}	31.0 _{2.7}	42.2 _{3.1}	51.6 _{2.9}	57.7 _{2.0}	73.2 _{0.0}	73.0 _{0.4}	77.4 _{1.3}	77.8 _{1.0}
MAD-M ²	90.3 _{3.5}	91.3 _{3.1}	81.9 _{2.2}	82.6 _{1.7}	48.0 _{3.2}	49.4 _{2.5}	52.4 _{2.4}	55.1 _{1.7}	72.7 _{1.7}	73.0 _{0.8}	77.0 _{0.7}	77.7 _{0.0}
Uncertain Prompt	95.3 _{1.2}	90.3 _{1.2}	89.0 _{0.9}	85.2 _{0.5}	38.3 _{2.4}	44.6 _{3.2}	55.8 _{3.3}	60.3 _{0.8}	72.2 _{1.1}	72.3 _{1.2}	77.0 _{1.5}	78.2 _{1.0}
Majority Vote	98.3 _{1.2}	98.3 _{1.2}	89.3 _{1.0}	89.3 _{1.0}	54.3 _{1.9}	54.3 _{1.9}	49.4 _{2.0}	49.4 _{2.0}	72.1 _{1.3}	72.1 _{1.3}	76.4 _{1.1}	76.4 _{1.1}
Vote Prompt	92.3 _{2.9}	91.0 _{3.5}	83.4 _{0.7}	80.1 _{1.5}	43.3 _{2.0}	46.2 _{6.3}	57.4 _{4.0}	52.6 _{1.7}	73.2 _{2.3}	73.2 _{1.9}	77.1 _{1.0}	77.6 _{1.0}
DAR (Ours)	96.7 _{0.6}	93.0 _{2.6}	89.9 _{0.4}	87.2 _{2.5}	41.1 _{1.9}	49.6 _{1.6}	58.2 _{3.8}	59.0 _{4.0}	71.6 _{1.8}	72.2 _{1.7}	77.9 _{0.8}	78.3 _{0.3}
Llama3.1-8B												
Single-agent	81.0 _{11.3}		72.3 _{1.4}		49.5 _{4.0}		45.6 _{3.9}		66.3 _{2.4}		63.7 _{1.9}	
Society Of Mind	91.5 _{0.7}	92.5 _{2.1}	85.4 _{0.2}	82.9 _{2.5}	53.2 _{2.5}	52.4 _{2.9}	48.1 _{2.4}	48.7 _{3.3}	66.5 _{1.8}	58.2 _{3.7}	67.9 _{2.8}	66.9 _{3.3}
MAD-M ²	74.3 _{16.3}	78.0 _{15.7}	60.7 _{4.5}	64.1 _{4.5}	52.0 _{1.0}	51.0 _{1.0}	35.2 _{1.7}	39.2 _{3.3}	66.7 _{2.3}	71.7 _{2.6}	67.6 _{1.6}	68.1 _{0.5}
Uncertain Prompt	95.0 _{1.4}	94.5 _{0.7}	83.0 _{0.9}	83.9 _{1.5}	52.1 _{0.8}	50.7 _{0.3}	45.6 _{1.7}	48.8 _{1.0}	67.9 _{2.8}	63.2 _{4.1}	68.7 _{0.3}	68.9 _{1.3}
Majority Vote	91.4 _{3.1}	91.4 _{3.1}	85.4 _{0.7}	85.4 _{0.7}	53.2 _{2.2}	53.2 _{2.2}	45.6 _{2.6}	45.6 _{2.6}	70.0 _{1.7}	70.0 _{1.7}	65.5 _{1.6}	65.5 _{1.6}
Vote Prompt	89.0 _{1.2}	91.0 _{0.7}	82.0 _{1.3}	81.0 _{2.9}	50.0 _{2.1}	52.0 _{1.8}	51.6 _{2.1}	51.3 _{2.3}	65.8 _{4.1}	57.1 _{5.2}	68.3 _{0.7}	66.2 _{3.2}
DAR (Ours)	95.5 _{2.1}	94.0 _{1.4}	85.2 _{1.6}	84.3 _{2.4}	54.1 _{1.6}	53.8 _{1.7}	49.2 _{2.4}	49.7 _{1.8}	71.2 _{2.2}	68.0 _{5.4}	70.4 _{0.7}	70.8 _{0.2}

Table 9: Performance of different methods across models and datasets (Decentralized MAD, $N=8$).

Method	Arithmetics		GSM8K		HH-RLHF		Form.Log.		Pro.Med.		CSQA	
	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2
Qwen2.5-1.5B												
Society Of Mind	83.3 _{11.0}	83.3 _{10.0}	72.5 _{1.2}	70.7 _{1.4}	39.2 _{6.8}	38.3 _{0.0}	37.0 _{4.7}	36.2 _{3.2}	57.2 _{2.0}	57.5 _{2.4}	64.4 _{2.0}	64.6 _{1.9}
MAD-M ²	44.7 _{23.6}	43.0 _{27.2}	59.8 _{3.2}	56.8 _{7.3}	51.3 _{4.1}	50.4 _{3.7}	39.7 _{1.6}	38.4 _{2.0}	57.7 _{1.3}	57.6 _{1.7}	64.2 _{2.5}	64.7 _{2.5}
Uncertain Prompt	84.0 _{3.6}	83.0 _{4.0}	73.8 _{4.9}	68.2 _{3.5}	31.5 _{0.2}	24.8 _{1.2}	30.7 _{1.2}	23.0 _{1.6}	56.9 _{2.2}	56.8 _{2.1}	62.8 _{2.9}	63.1 _{2.6}
Majority Vote	80.4 _{4.0}	80.4 _{4.0}	69.5 _{1.2}	69.5 _{1.2}	42.4 _{2.3}	42.4 _{2.3}	32.7 _{3.4}	32.7 _{3.4}	58.7 _{3.1}	58.7 _{3.1}	68.6 _{0.6}	68.6 _{0.6}
Vote Prompt	85.0 _{2.6}	87.3 _{4.0}	72.3 _{1.2}	70.7 _{0.8}	36.7 _{0.9}	37.5 _{3.1}	37.0 _{0.5}	34.4 _{5.4}	58.1 _{3.4}	58.2 _{3.5}	63.1 _{2.5}	62.9 _{2.4}
DAR (Ours)	85.0 _{2.8}	88.0 _{4.2}	75.0 _{0.5}	72.5 _{0.7}	43.8 _{0.7}	43.8 _{1.6}	38.4 _{2.6}	36.2 _{5.1}	58.2 _{3.1}	57.8 _{3.1}	68.4 _{4.2}	68.3 _{3.6}
Qwen2.5-3B												
Society Of Mind	66.3 _{3.8}	59.0 _{6.1}	73.6 _{1.7}	66.8 _{2.3}	49.3 _{1.7}	46.8 _{3.2}	43.7 _{2.9}	44.4 _{1.4}	63.1 _{1.5}	62.4 _{1.5}	72.8 _{2.1}	73.1 _{1.6}
MAD-M ²	69.7 _{8.1}	67.7 _{9.9}	66.0 _{5.9}	65.7 _{3.1}	40.0 _{1.7}	39.9 _{3.8}	31.3 _{1.4}	31.3 _{1.4}	63.2 _{1.3}	63.2 _{1.1}	71.2 _{1.6}	71.7 _{2.1}
Uncertain Prompt	67.0 _{5.6}	57.0 _{4.6}	75.3 _{0.5}	66.0 _{1.4}	48.8 _{2.2}	44.4 _{1.2}	33.3 _{2.1}	26.5 _{2.3}	63.6 _{1.0}	63.4 _{1.1}	72.4 _{1.2}	71.8 _{1.2}
Majority Vote	78.6 _{2.8}	78.6 _{2.8}	83.2 _{1.0}	83.2 _{1.0}	49.7 _{1.5}	49.7 _{1.5}	43.8 _{3.1}	43.8 _{3.1}	63.8 _{1.4}	63.8 _{1.4}	72.2 _{0.9}	72.2 _{0.9}
Vote Prompt	76.3 _{2.1}	65.7 _{2.5}	75.3 _{0.5}	66.0 _{1.4}	47.7 _{2.2}	46.2 _{2.5}	46.3 _{4.8}	42.3 _{1.2}	63.7 _{2.2}	63.2 _{1.9}	71.6 _{0.5}	71.9 _{0.7}
DAR (Ours)	85.0 _{2.6}	79.3 _{2.9}	86.3 _{0.9}	82.0 _{0.0}	49.2 _{0.2}	50.8 _{2.6}	47.6 _{2.9}	44.7 _{2.8}	64.7 _{1.9}	64.4 _{0.8}	73.2 _{1.0}	73.3 _{1.3}
Falcon3-7B												
Society Of Mind	98.0 _{1.7}	99.3 _{0.6}	90.6 _{0.5}	90.7 _{1.4}	45.7 _{2.6}	48.7 _{1.8}	64.6 _{2.0}	64.6 _{2.4}	71.7 _{1.3}	72.5 _{0.8}	76.7 _{1.2}	76.6 _{1.1}
MAD-M ²	89.7 _{3.1}	90.0 _{0.0}	81.7 _{2.2}	81.6 _{1.8}	48.2 _{3.3}	51.3 _{3.8}	55.0 _{1.2}	56.3 _{1.4}	72.3 _{0.8}	72.8 _{1.5}	77.0 _{0.6}	77.1 _{0.8}
Uncertain Prompt	100.0 _{0.0}	99.3 _{0.6}	93.1 _{0.8}	90.7 _{0.9}	51.0 _{3.5}	49.7 _{3.9}	62.7 _{0.8}	64.8 _{2.4}	71.7 _{0.4}	72.4 _{1.0}	76.1 _{0.8}	76.2 _{0.8}
Majority Vote	99.8 _{0.2}	99.8 _{0.2}	91.7 _{1.0}	91.7 _{1.0}	54.5 _{2.9}	54.5 _{2.9}	52.8 _{2.3}	52.8 _{2.3}	71.8 _{1.2}	71.8 _{1.2}	76.0 _{1.3}	76.0 _{1.3}
Vote Prompt	98.7 _{1.5}	98.3 _{0.6}	92.2 _{0.2}	90.3 _{0.9}	52.7 _{2.8}	50.0 _{3.3}	61.6 _{3.0}	59.0 _{3.2}	72.2 _{0.8}	72.3 _{0.6}	77.2 _{0.7}	77.6 _{0.7}
DAR (Ours)	99.7 _{0.6}	100.0 _{0.0}	93.0 _{0.9}	91.1 _{0.5}	53.2 _{2.3}	52.4 _{1.5}	61.4 _{1.7}	66.1 _{1.2}	72.7 _{0.2}	73.4 _{0.9}	76.6 _{0.2}	77.7 _{0.6}
Llama3.1-8B												
Society Of Mind	96.0 _{1.0}	95.7 _{1.5}	88.7 _{0.9}	86.5 _{2.6}	54.8 _{4.0}	53.0 _{2.4}	51.3 _{2.0}	53.7 _{2.4}	70.8 _{2.4}	65.7 _{0.2}	69.7 _{2.4}	69.3 _{3.6}
MAD-M ²	70.0 _{15.1}	73.7 _{17.9}	60.2 _{4.0}	61.4 _{3.1}	51.8 _{0.7}	50.8 _{0.8}	33.6 _{2.4}	35.7 _{1.6}	68.3 _{1.8}	59.4 _{5.8}	69.8 _{2.7}	68.2 _{3.8}
Uncertain Prompt	97.0 _{1.0}	96.3 _{0.6}	88.0 _{0.9}	86.5 _{0.7}	53.2 _{4.0}	52.7 _{3.8}	53.2 _{0.8}	51.9 _{2.4}	70.8 _{3.5}	66.8 _{4.7}	69.0 _{1.7}	68.0 _{2.2}
Majority Vote	97.2 _{0.7}	97.2 _{0.7}	88.9 _{1.4}	88.9 _{1.4}	53.3 _{4.3}	53.3 _{4.3}	49.6 _{3.0}	49.6 _{3.0}	74.9 _{1.5}	74.9 _{1.5}	67.6 _{1.4}	67.6 _{1.4}
Vote Prompt	96.3 _{1.2}	93.3 _{0.6}	88.8 _{1.2}	85.3 _{0.9}	55.0 _{0.3}	51.7 _{1.2}	54.2 _{2.0}	53.4 _{1.2}	68.5 _{1.5}	61.0 _{4.1}	68.7 _{2.6}	67.8 _{2.0}
DAR (Ours)	97.3 _{2.1}	96.7 _{0.6}	89.7 _{0.0}	87.8 _{0.7}	55.3 _{0.3}	54.0 _{0.0}	55.6 _{2.7}	53.2 _{1.4}	75.9 _{1.7}	72.0 _{2.0}	71.0 _{1.5}	70.6 _{1.1}

Table 10: Performance of different methods across models and datasets (Sparse MAD).

Method	Arithmetics		GSM8K		HH-RLHF		Form.Log.		Pro.Med.		CSQA	
	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2	R=1	R=2
Qwen2.5-1.5B												
Single-agent	42.0 _{7.1}		38.2 _{2.1}		26.0 _{1.9}		24.2 _{0.6}		55.6 _{2.8}		60.4 _{1.7}	
Society Of Mind	63.5 _{0.7}	72.0 _{4.2}	63.0 _{1.4}	62.7 _{4.2}	33.3 _{1.4}	33.2 _{1.6}	32.5 _{4.5}	33.7 _{3.9}	57.8 _{1.9}	57.6 _{1.9}	63.9 _{2.5}	64.4 _{2.2}
MAD-M ²	44.0 _{24.8}	37.7 _{26.4}	59.3 _{2.6}	50.7 _{9.6}	50.4 _{4.7}	50.9 _{2.3}	34.7 _{0.8}	34.9 _{0.5}	57.7 _{1.7}	57.5 _{1.9}	63.9 _{2.5}	64.3 _{2.4}
Uncertain Prompt	67.5 _{4.9}	60.5 _{9.2}	61.5 _{2.6}	57.7 _{5.2}	34.0 _{2.4}	31.5 _{0.7}	35.3 _{3.9}	31.7 _{4.5}	57.4 _{2.2}	57.6 _{2.0}	63.7 _{2.6}	64.0 _{2.6}
Majority Vote	59.2 _{3.5}	59.2 _{3.5}	60.4 _{0.1}	60.4 _{0.1}	37.2 _{0.5}	37.2 _{0.5}	30.5 _{1.7}	30.5 _{1.7}	58.6 _{2.9}	58.6 _{2.9}	68.7 _{0.7}	68.7 _{0.7}
Vote Prompt	68.5 _{0.7}	72.0 _{1.4}	62.8 _{0.2}	59.2 _{4.9}	34.3 _{0.8}	35.0 _{3.3}	36.9 _{1.7}	37.7 _{1.7}	57.8 _{2.5}	57.7 _{2.7}	64.4 _{2.0}	64.6 _{2.0}
DAR (Ours)	70.5 _{0.7}	75.0 _{5.7}	64.0 _{1.9}	58.8 _{4.0}	33.5 _{6.4}	38.0 _{1.9}	37.3 _{1.1}	33.3 _{3.4}	59.1 _{3.0}	59.1 _{3.4}	66.6 _{3.0}	66.2 _{3.0}
Qwen2.5-3B												
Single-agent	43.5 _{0.7}		34.7 _{0.9}		40.2 _{3.1}		28.6 _{2.2}		59.8 _{1.6}		69.2 _{2.7}	
Society Of Mind	63.0 _{2.8}	52.5 _{7.8}	68.5 _{1.2}	60.0 _{1.4}	46.2 _{0.7}	43.2 _{1.6}	39.3 _{0.6}	37.3 _{2.2}	64.5 _{1.7}	64.1 _{1.5}	71.4 _{1.7}	72.0 _{0.9}
MAD-M ²	69.0 _{10.4}	66.7 _{10.0}	66.0 _{0.9}	60.1 _{3.3}	39.0 _{4.9}	38.2 _{4.7}	33.1 _{3.2}	42.3 _{2.3}	64.8 _{2.0}	66.3 _{2.5}	71.2 _{0.8}	71.3 _{1.2}
Uncertain Prompt	65.0 _{5.7}	53.0 _{4.2}	69.0 _{2.8}	54.3 _{0.5}	44.2 _{4.9}	41.7 _{1.9}	42.5 _{1.7}	40.1 _{0.6}	64.1 _{1.7}	63.7 _{1.3}	70.2 _{0.7}	70.3 _{1.2}
Majority Vote	65.8 _{4.5}	65.8 _{4.5}	70.2 _{1.9}	70.2 _{1.9}	45.7 _{1.1}	45.7 _{1.1}	42.3 _{1.1}	42.3 _{1.1}	63.7 _{1.0}	63.7 _{1.0}	71.3 _{0.9}	71.3 _{0.9}
Vote Prompt	70.0 _{4.2}	54.0 _{5.7}	67.2 _{0.7}	55.2 _{0.2}	45.3 _{2.8}	42.7 _{0.9}	39.7 _{1.1}	40.9 _{2.8}	64.8 _{0.6}	63.8 _{0.8}	70.6 _{1.7}	70.8 _{0.8}
DAR (Ours)	70.5 _{4.9}	61.5 _{3.5}	70.8 _{2.6}	54.7 _{2.4}	48.3 _{0.9}	44.2 _{1.2}	40.9 _{3.9}	35.7 _{3.4}	65.6 _{1.1}	65.7 _{0.4}	71.8 _{1.0}	71.2 _{1.3}
Falcon3-7B												
Single-agent	58.0 _{1.4}		54.8 _{0.7}		37.0 _{1.4}		40.9 _{0.6}		70.3 _{0.4}		75.0 _{0.9}	
Society Of Mind	84.5 _{0.7}	90.0 _{2.8}	81.0 _{2.8}	76.8 _{4.9}	34.5 _{0.2}	43.7 _{2.8}	53.6 _{0.6}	57.5 _{0.6}	71.0 _{1.3}	71.9 _{0.6}	77.2 _{1.3}	77.8 _{1.3}
MAD-M ²	90.3 _{3.2}	90.7 _{2.5}	81.7 _{2.2}	81.6 _{1.8}	49.1 _{4.8}	52.1 _{4.0}	51.1 _{0.0}	58.7 _{1.4}	72.2 _{1.3}	72.8 _{1.0}	77.3 _{1.5}	78.1 _{1.5}
Uncertain Prompt	95.0 _{0.0}	85.5 _{0.7}	86.7 _{3.3}	78.0 _{1.4}	43.0 _{4.2}	47.5 _{3.1}	57.5 _{2.8}	54.8 _{4.5}	71.9 _{1.5}	72.3 _{1.7}	77.2 _{0.2}	77.6 _{0.5}
Majority Vote	97.2 _{2.1}	97.2 _{2.1}	88.0 _{1.4}	88.0 _{1.4}	53.0 _{2.8}	53.0 _{2.8}	48.7 _{3.4}	48.7 _{3.4}	71.6 _{0.9}	71.6 _{0.9}	76.7 _{1.3}	76.7 _{1.3}
Vote Prompt	91.5 _{2.1}	89.5 _{0.7}	87.0 _{3.3}	79.8 _{5.4}	42.2 _{1.2}	46.5 _{2.1}	57.1 _{3.4}	57.5 _{0.6}	72.7 _{0.8}	72.7 _{0.8}	77.0 _{1.0}	77.6 _{0.8}
DAR (Ours)	95.0 _{2.8}	91.0 _{2.8}	89.7 _{1.9}	82.0 _{0.0}	40.7 _{1.9}	47.0 _{2.8}	60.3 _{6.7}	57.9 _{3.4}	72.2 _{0.8}	72.8 _{0.4}	77.4 _{0.5}	77.8 _{0.8}
Llama3.1-8B												
Single-agent	79.0 _{2.8}		74.0 _{0.9}		49.0 _{0.9}		47.2 _{9.5}		66.3 _{2.4}		63.7 _{1.9}	
Society Of Mind	95.5 _{2.1}	93.0 _{0.0}	85.5 _{2.1}	84.8 _{1.6}	54.0 _{2.4}	52.3 _{6.1}	49.2 _{1.1}	50.0 _{1.1}	66.7 _{6.2}	63.0 _{5.5}	68.4 _{1.0}	67.9 _{1.3}
MAD-M ²	76.3 _{15.9}	80.0 _{16.5}	55.5 _{2.6}	61.0 _{1.6}	50.0 _{1.2}	51.3 _{1.2}	34.1 _{3.6}	34.7 _{3.7}	66.1 _{3.1}	70.1 _{5.3}	67.2 _{1.2}	68.2 _{2.3}
Uncertain Prompt	93.0 _{5.7}	93.0 _{2.8}	83.7 _{0.5}	83.7 _{1.4}	52.3 _{0.0}	51.2 _{0.7}	52.4 _{4.5}	49.2 _{2.2}	70.6 _{3.3}	67.9 _{3.8}	68.7 _{2.1}	68.2 _{1.6}
Majority Vote	90.7 _{4.2}	90.7 _{4.2}	84.5 _{1.1}	84.5 _{1.1}	52.0 _{1.3}	52.0 _{1.3}	45.5 _{4.3}	45.5 _{4.3}	74.6 _{1.5}	74.6 _{1.5}	65.4 _{2.0}	65.4 _{2.0}
Vote Prompt	93.5 _{2.1}	94.5 _{2.1}	85.0 _{1.9}	82.7 _{1.4}	51.0 _{2.5}	47.0 _{2.2}	52.4 _{4.5}	49.2 _{2.2}	67.9 _{1.9}	62.4 _{3.2}	67.8 _{0.2}	67.3 _{1.5}
DAR (Ours)	94.5 _{0.7}	94.5 _{2.1}	86.3 _{0.8}	84.0 _{1.5}	54.2 _{3.5}	50.3 _{1.9}	52.0 _{0.6}	51.6 _{0.0}	72.2 _{2.1}	67.2 _{4.1}	67.6 _{2.8}	68.7 _{1.8}

Table 11: Wall-clock time per debate round (in second) on Arithmetics and Formal Logic tasks ($N=4$). Values for our method include $\times$, indicating the ratio relative to Society Of Mind.

Model	Dataset	Society Of Mind	DAR (Ours)
Qwen2.5-1.5B	Arithmetics	51 $\pm$ 5	58 $\pm$ 7 ($\times$ 1.14)
	Form.Log.	45 $\pm$ 1	51 $\pm$ 3 ($\times$ 1.13)
Qwen2.5-3B	Arithmetics	82 $\pm$ 11	85 $\pm$ 4 ($\times$ 1.03)
	Form.Log.	108 $\pm$ 14	113 $\pm$ 3 ($\times$ 1.05)

Question: What is the result of $27 + 6 \times 15 + 7 - 0 \div 22$?

Correct Answer: 124

Majority Vote (Round 0)

Agent 1: Here's a formula to guide you through it:

(first number) + (second number * third number) - fourth number / fifth number. To solve this problem, we need to follow the order of operations, also known as PEMDAS (Parentheses, Exponents, Multiplication and Division from left to right, Addition and Subtraction from left to right). In this case, there are no parentheses or exponents, so we will only do multiplication, division, addition, and subtraction.

So, let's break down the expression step by step:

$\{27 + 6(15) + 7 - 0/22\}$

First, we do the multiplication: $\{27 + 90 + 7 - 0/22\}$

Next, we perform any division: $\{27 + 90 + 7 - 0\}$

Then, we perform the addition and subtraction from left to right: $\{117 - 0\} = \{117\}$

Therefore, the final answer is: "{final answer: 117}"

Agent 2: Your final answer should match the highest probability of being correct based on the agent's responses.

The curly braces {} enclose text, not numbers. So if you get a result of 285, your final answer should look like this: "{final answer: 285}"

You may use multiple words such as "final answer:" if desired. The key is to present a single string of text rather than individual characters inside curly braces. You will lose points if your submission does not match this exact format. Do not submit additional explanations apart from what is already stated within the curly braces. Avoid repetition in final submission. Focus solely on providing the final answer enclosed in curly braces. Based on the available opinions, what is the result of $27 + 6 * 15 + 7 - 0 / 22$?

Final answer: {117}.

Agent 3: The expression you've provided has parentheses for grouping operations. According to standard order of operations (PEMDAS/BODMAS), multiplication and division are performed before addition and subtraction.

Let's break it down step by step:

1. First, perform any multiplications and divisions from left to right:

- $(6 \times 15 = 90) - (0 / 22 = 0)$

So now our equation looks like this after these operations:

$\{27 + 90 + 7 - 0\}$

2. Now we can add/subtract from left to right:

- Add $(27 + 90 = 117)$

- Then subtract $(117 + 7 = 124)$

- Finally, there's no other operation left on the number line to remove, so we're done here. Therefore, the final answer is: "{final answer: 124}."

Final: 117 (X)

Figure 13: Complete qualitative responses on Qwen2.5-1.5B (Majority Vote).

Question: What is the result of $27 + 6 \times 15 + 7 - 0 \div 22$?
Correct Answer: 124

Round 1

Agent 1: Here's a tricky step-by-step explanation: We start with the given expression: $\{27 + 6 \times 15 + 7 - 0 / 22\}$.

The first operation we encounter is multiplication and division from left to right. First, we handle multiplication and division:

$\{27 + (6 \times 15) + 7 - (0 / 22)\}$

Since there are no parentheses or other grouping symbols that require us to perform operations inside them first, this simplifies down to:

$\{27 + 90 + 7 - (0 / 22)\}$

We know that dividing by zero isn't allowed in mathematics because it leads to undefined results, so let's deal with what remains after that part:

$\{27 + 90 + 7 - 0\}$

Since any number divided by itself equals one, we get:

$\{117 + 0\}$

Finally, adding everything up gives us our final sum: $\{117\}$

Thus, the solution can be stated as "{final answer: 117}"

Agent 2: To solve this problem correctly following the order of operations (PEMDAS/BODMAS), we need to perform multiplication before addition and subtraction.

Here's how it goes:

$\{27 + 6 \times 15 + 7 - 0 / 22\} = \{27 + 90 + 7 - 0\} = \{117\}$

So the final answer in curly brackets is: "{final answer: 117}"

Note that division by zero is undefined, which has been accounted for in the calculation since there was a term with $0/22$. If you had an expression without fractions or divisions by zero, such as $12 \times 6 + 1$, where the result would be 73, the process could follow these steps: First multiply 12×6 giving us 72, then add 1 to get our final result of 73. In cases where you have mixed numbers, such as $12 \frac{1}{2}$ or " a/b ", the multiplication should be performed on the whole number and the fractional part separately before summing them up. For example, if you have the expression $12 \frac{1}{2} + 4/2$, first do $(12 \times 1) + (1 \times 2) = 14$ then $(4 \times 2) / 2 = 4$, so the final result would be $14 + 4 = 18$. Without further details on what kind of arithmetic operation might be involved beyond just simple additions/subtractions within the given parentheses, no more calculations can be made. If you meant something else entirely by the question, please clarify the context! However, based purely on the numerical values presented here, the correct final answer to the equation given is simply "117".

Agent 3: To solve the expression

$(27 + 6 \times 15 + 7 - 0 \div 22)$, we should follow the order of operations (PEMDAS/BODMAS rules):

1. First, calculate any multiplication and division from left to right: $6 \times 15 = 90$. Now the expression becomes: $27 + 90 + 7 - 0 \div 22$

2. Next, perform the division: $0 \div 22 = 0$. Now the expression simplifies further: $27 + 90 + 7 - 0$

3. Finally, add and subtract from left to right: $27 + 90 = 117$. Then, $117 + 7 = 124$. And finally, $124 - 0 = 124$.

The final answer is (final answer: 124).

Round 2

Agent 1: To solve this expression step by step while correctly applying the order of operations (PEMDAS), let's break it down into parts:

First, we perform the multiplication and division: $\{6 \times 15 + 0 / 22\} \implies \{90 + 0\} = 90$

Now, move on to addition and subtraction: $\{27 + 90 - 0\}$

Finally, calculate: $\{27 + 90 - 0\} = 117 - 0 \{117 - 0\} = 117$

Therefore, the final answer is:

"{final answer: 117}"

Agent 2: Use markdown for formatting your response properly.

If you have no new opinions and want to stop commenting, simply write "nothing new" and press enter. You won't receive another agent message from me until you add some comments.

Do not reply to the following prompt: Nothing new. Assistant: {final answer: 117}

I see everything worked out correctly according to my calculations. This matches the manual computations we did earlier, confirming our approach and ensuring accuracy in each step. Your verification helps solidify our conclusion, which is indeed "{final answer: 117}"

Agent 3: To solve this problem, we need to follow the order of operations, commonly remembered by the acronym PEMDAS (Parentheses, Exponents, Multiplication and Division from left to right, Addition and Subtraction from left to right).

Here's the given expression: $(27 + 6 \times 15 + 7 - 0 / 22)$

Step-by-step solution:

1. **Multiplication**: $(6 \times 15 = 90)$

2. **Division**: $(0 / 22 = 0)$ because any number divided by zero is undefined or considered as zero.

Now our expression looks like this after performing the multiplication and division: $(27 + 90 + 7 - 0)$

3. Perform addition and subtraction from left to right: $(27 + 90 = 117)$, $(117 + 7 = 124)$, $(124 - 0 = 124)$.

So, the final result is (124). Therefore, the final answer is: "{final answer: 124}"

Final: 117 (X)

Figure 14: Complete qualitative responses on Qwen2.5-1.5B (Society Of Mind).

Question: What is the result of $27 + 6 \times 15 + 7 - 0 \div 22$?
Correct Answer: 124

Round 1

Agent 1: Based on the latest opinions, what is the result of $27 + 6 \times 15 + 7 - 0 \div 22$?

Final answer: {117} Uncertainty score: 0.7183

Based on the uncertainty scores and the majority response, the best guess for the final answer appears to be 117. This approach ensures clarity and adherence to the instruction for enclosing the final answer in curly braces while acknowledging the confidence level among the agents regarding the accuracy.

The majority vote answer from the previous analysis suggests the result might be 285. However, considering factors such as precision, logical reasoning, and rounding, I believe the closest and more accurate result based on the options given is 117. Thus, my final answer is encapsulated within curly braces: {final answer: 117} with an uncertainty score of 0.7183.

Agent 2: Use your knowledge of the correct sequence of operations in order to obtain the desired final result. Here is a table summarizing the contributions of the experts: - Agent A: "I interpreted the operation order differently. If PEMDAS applies, then $27+6 \times 15=90$ after multiplying and $90+7$ gives us 97. Then $97-0$ gives us 97"

- Agent B: "My logic goes something like this: 6×15 gives us 90. If 0 is divided by any non-zero number (like 22), it still equals 0. Hence, it is okay not to include it in the overall operation.

- Agent C: "Since division is not explicitly mentioned as being a part of the PEMDAS rule, I followed standard procedures."

Using the insights from Agents B and C's contributions, and **incorporating the majority vote**, the best approach seems to be taking the standard approach of evaluating the multiplication and division operations first (since PEMDAS prioritizes multiplication and division over addition and subtraction). Given that 0 divided by any non-zero number remains 0, we disregard it during our calculation and proceed directly with the standard PEMDAS sequence: multiplication and division come before addition and subtraction. So our steps would be: ($6 \times 15 = 90$). Adding subsequent terms: ($90 + 7 = 97$). Substituting back: ($97 - 0 = 97$). However, since division by zero is undefined, we exclude it, thus the final result is merely ($27 + 6 \times 15 + 7 = 97$). The key insight is avoiding direct consideration of ($0 \div 22$).

Therefore, following the standard order of operations and excluding division by zero where applicable, the final result is: {final answer: 124}.

Agent 3: Use the following formula for conversion: {final answer: }. Include your reasoning in your reply to receive full credit. In my previous attempt, I misinterpreted the order of operations and overlooked division by zero. Thank you for correcting my understanding and guiding me on the right path again. Your expertise has been invaluable. Based on the clarified instructions and the unanimous agreement from all experts, I must admit that **the majority's perspective led to an entirely different calculation method and answer.** Following the correct order of operations (PEMDAS/BODMAS) and applying standard procedures without including division by zero:

1. Begin with (6×15): ($6 \times 15 = 90$)

2. Next, evaluate ($0 \div 22$): ($0 \div 22 = 0$)

3. Substitute back into the expression: ($27 + 90 + 7 - 0$)

4. Perform the addition and subtraction: ($27 + 90 = 117$), ($117 + 7 = 124$), ($124 - 0 = 124$).

Thus, following proper order of operations and eliminating the undefined case of ($0 \div 22$), the final result is: {final answer: 124}.

Round 2

Agent 1: Here is your recent opinion: {final answer: 117}. Please show all calculations and explanations.

Your final answer should match the higher chance of the agent's answers matching the result. Ensure you also show how you arrived at the answer in detail and provide a comprehensive explanation including the reasoning for the final calculation. Thank you! To find the result of the expression ($27 + 6 \times 15 + 7 - 0 \div 22$), we need to follow the order of operations (PEMDAS/BODMAS), which stands for Parentheses/Brackets, Exponents/Orders, Multiplication and Division (from left to right), and Addition and Subtraction (from left to right). Let's go through each step in detail.

Step-by-Step Solution:

1. Multiplication and Division

First, handle the multiplication and division: ($6 \times 15 = 90$) - ($0 \div 22 = 0$). So, the expression simplifies to: ($27 + 90 + 7 - 0$)

2. Addition and Subtraction

Next, add the values together: ($27 + 90 = 117$), ($117 + 7 = 124$). Now, we only need to perform the final subtraction: ($124 - 0 = 124$)

Detailed Explanation:

The expression involves two types of operations: multiplication (6×15) and division ($0 \div 22$). According to the order of operations (PEMDAS/BODMAS), we always perform multiplication and division from left to right before addition and subtraction. However, since any value divided by itself equals one (for example, ($0 \div 0 = 1$), division by zero essentially "cancels out," making it equivalent to zero.

Thus, we effectively ignore the division term because adding a value to zero does not affect the sum. This is a crucial aspect of correctly applying the order of operations in algebraic expressions.

Final Result:

After performing the multiplication and division operations correctly (ignoring division by zero), we arrive at: {124}.

Agent 2: "Consider the possible reasons behind the disagreement, including whether division by zero is defined as zero, and how it affects the overall calculation. Based on all the given information and your own reasoning, please provide a final answer. The answer can only contain numbers and/or decimal points. There must not be any spaces between them. Here is your starting point for revising the previous responses, make any changes needed to comply with the new guidelines:

You calculated ($27 + 6 \times 15 + 7 - 0 \div 22$) incorrectly according to PEMDAS and did not take into account the potential impact of division by zero, hence arriving at the wrong answer. Using the correct PEMDAS/BODMAS rule for evaluation ensures accurate calculation:

1. Multiply (6×15): ($6 \times 15 = 90$)

2. Add the next term, (27): ($27 + 90 = 117$)

3. Add (7): ($117 + 7 = 124$)

4. Subtract ($0 \div 22$): ($0 \div 22 = 0$)

5. Finally, subtract (0) from (124): ($124 - 0 = 124$)

Hence, the correct final answer is: {final answer: 124}.

Agent 3: Additionally, consider your uncertainty score and provide a detailed explanation of how you arrived at your answer, including any mistakes or areas of confusion you encountered during your calculations. Lastly, thank everyone for their feedback and input. {final answer: 124} with an uncertainty score of 0.5. Thank you for sharing your detailed thoughts and calculations. After carefully reviewing your comprehensive explanation and the various approaches taken, here's a refined final answer considering the nuances in interpretation: {final answer: 124}

My analysis reveals a series of logical steps that led to this result, emphasizing clear adherence to the standard order of operations (PEMDAS). Specifically:

1. ****Multiplication****: ($6 \times 15 = 90$)

2. ****Addition & Subtraction****: Begin by performing the remaining multiplications and divisions next, since they come before addition/subtraction: ($0 \div 22 = 0$). Thus, ($27 + 90 + 7 - 0$)

3. ****Performing Addition and Subtraction****: ($27 + 90 = 117$), ($117 + 7 = 124$)

By proceeding through these steps and confirming the absence of undefined division operations, we arrive at a precise final sum of ****124****.

Considering possible interpretational variances, such as differing emphases on whether zero is undefined or can be treated as zero. We have maintained a moderate uncertainty score of 0.5, acknowledging the balance of multiple perspectives and the critical role of precise, standardized operations. For future clarity and uniformity, please use the notation {final answer: 124}, ensuring consistency across all entries.

Final: 124 (✓)

Figure 15: Complete qualitative responses on Qwen2.5-1.5B (DAR).